Week 1 : Assignment 1

Due date on 2026-08-05, 23:59 IST

The due date for submitting this assignment has passed.

Your last recorded submission was on 2026-07-13, 20:38 IST.

Case Study: Programming Beyond Coding, Why Every Professional Can Benefit

Riya is a first-year university student pursuing a degree in Economics. She believes learning programming is unnecessary because there are already apps and websites for almost every task. During an orientation session, her professor explains that the existence of many applications does not eliminate the need for programming. Just as existing buildings do not prevent people from constructing or improving new ones, existing software can always be enhanced, customized, or automated to solve new problems.

The professor further explains that programming is not limited to computer science. Whether someone studies mathematics, physics, economics, business, or nutrition, programming helps automate repetitive tasks, analyze data, and build customized solutions that are not available in ready-made software.

When students ask why there are so many programming languages, the professor explains that different languages evolved to solve different kinds of problems. Languages such as FORTRAN focused on scientific computing, COBOL on business applications, C emphasized efficiency, C++ introduced object-oriented programming, and Python made programming easier and more accessible. As technology evolves, programming languages and development tools continue to become simpler and more powerful.

To help students understand what programming really means, the professor conducts a classroom activity. He assigns different hand gestures to represent coffee, tea, water, juice, and cookies. Students quickly learn to interpret combinations of gestures, such as "three cups of coffee" or "two cookies." The professor explains that this activity is similar to programming: a programmer gives precise instructions using symbols that a computer has been taught to understand.

By the end of the session, Riya realizes that programming is not just about writing code, it is a problem-solving skill that can provide an advantage in almost any profession.

1.

Which statement best captures the professor's reason for learning programming despite the availability of many applications?

1 Point

Existing applications eliminate the need for programmers.

Programming enables people to create, improve, and customize solutions beyond existing applications.
Programming is useful only for developing mobile apps.
Learning programming is compulsory for every profession.

Yes, the answer is correct. Score: 1

Accepted answer isProgramming enables people to create, improve, and customize solutions beyond existing applications.

2.

According to the professor, why have multiple programming languages evolved over time?

1 Point

Each country develops its own programming language.
Different languages were designed to solve different types of problems and improve programming.
Older programming languages cannot run on modern computers.
Every programmer must create a new programming language.

Yes, the answer is correct. Score: 1

Accepted answer isDifferent languages were designed to solve different types of problems and improve programming.

3.

What was the primary purpose of the hand-gesture classroom activity?

1 Point

To improve students' memory.
To demonstrate that programming involves giving clear instructions using agreed-upon symbols.
To teach sign language.
To measure students' reaction speed.

Yes, the answer is correct. Score: 1

Accepted answer isTo demonstrate that programming involves giving clear instructions using agreed-upon symbols.

4.

According to the professor, what is the most effective way to become proficient in programming?

1 Point

Memorize every programming language before writing code.
Begin with advanced programming problems.
Practice regularly, stay motivated, and learn gradually through small steps.
Depend entirely on modern programming tools and IDEs.

Yes, the answer is correct. Score: 1

Accepted answer isPractice regularly, stay motivated, and learn gradually through small steps.

5.

Which conclusion would Riya most likely draw after attending the session?

1 Point

Programming is a problem-solving skill that can benefit people across many disciplines.
Programming is valuable only for software engineers.

Programming is primarily about memorizing the syntax of different languages.
Existing software packages make programming unnecessary.

Yes, the answer is correct. Score: 1

Accepted answer isProgramming is a problem-solving skill that can benefit people across many disciplines.

Case Study: Teaching a Robot Efficiently – The Power of Loops

A robotics club is programming a small robot to train for a fitness challenge. The robot must move forward and then return to its starting position. On every round, it should travel a little farther than before. Initially, one student writes separate instructions for every movement: move forward, pause, move backward, pause, and repeats this pattern many times with different distances.

Another student notices that the instructions follow the same pattern repeatedly. Instead of writing dozens of similar commands, she suggests using a loop along with a variable that stores the current distance. After every round, the variable is increased by a fixed amount, and the loop repeats until all rounds are completed.

The instructor points out that this shorter program is easier to understand, modify, and reuse. Later, instead of fixing the starting distance in the program, they introduce another variable so that the same program can be used for different fitness challenges without changing the main logic.

The students realize that loops are not just about reducing the number of lines of code—they make programs flexible, easier to maintain, and less prone to errors.

6.

Why did the second student recommend using a loop instead of writing every movement instruction separately?

1 Point

Loops allow repeated actions to be expressed with fewer instructions while preserving the same behavior.
Loops make the robot move faster.
Loops reduce the distance travelled by the robot.
Loops automatically correct programming errors.

Yes, the answer is correct. Score: 1

Accepted answer isLoops allow repeated actions to be expressed with fewer instructions while preserving the same behavior.

7.

In the improved program, what is the primary purpose of using a variable for the travel distance?

1 Point

To store the robot's battery level.
To allow the distance to change automatically after each repetition.
To count how many students are programming.
To increase the robot's speed.

Yes, the answer is correct. Score: 1

Accepted answer isTo allow the distance to change automatically after each repetition.

8.

The instructor later replaces the fixed starting distance with another variable. What is the main advantage of this change?

1 Point

The robot no longer requires a loop.
The same program can easily be reused for different starting distances without rewriting the logic.
The program will execute only once.
Variables automatically make the program run faster.

Yes, the answer is correct. Score: 1

Accepted answer isThe same program can easily be reused for different starting distances without rewriting the logic.

9.

Suppose a student removes the statement that updates the distance variable inside the loop. Which outcome is most likely?

1 Point

The robot will continue repeating the same movement distance in every round.
The robot will stop after the first movement.
The robot will automatically choose new distances.
The loop will immediately terminate.

Yes, the answer is correct. Score: 1

Accepted answer isThe robot will continue repeating the same movement distance in every round.

10.

Which statement best summarizes the role of loops in programming?

1 Point

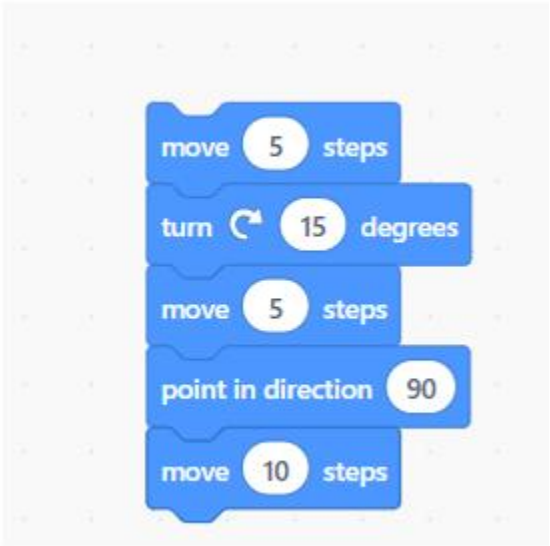
Loops are useful only for mathematical calculations.
Loops replace the need for variables.
Loops help automate repetitive tasks and make programs easier to modify and maintain.
Loops should be avoided because they make programs difficult to understand.

Yes, the answer is correct. Score: 1

Accepted answer isLoops help automate repetitive tasks and make programs easier to modify and maintain.

Case Study: Understanding a Scratch Program

A student is learning Scratch and creates the program shown in the figure below. The sprite starts from its initial position and executes each block from top to bottom exactly once. The student expects the sprite to follow the path specified by the movement and direction blocks. Study the Scratch program carefully that is presented below and answer the following questions based on the program shown.



11.

Which Scratch block determines the direction of the sprite before its final movement of 10 steps?

1 Point

move 5 steps (first block)
turn 15 degrees
move 5 steps (third block)
point in direction 90°

Yes, the answer is correct. Score: 1

Accepted answer ispoint in direction 90°

12.

What is the purpose of the turn 15 degrees block in this program?

1 Point

It changes the sprite's direction before the second movement.
It changes the sprite's position by 15 steps.
It changes the direction for the entire program, including after the point in direction block.
It makes the sprite move automatically.

Yes, the answer is correct. Score: 1

Accepted answer isIt changes the sprite's direction before the second movement.

13.

Suppose the point in direction 90° block is removed while keeping all other blocks unchanged. Which statement is most likely to be true?

1 Point
The final movement will continue in the direction established by the earlier turn.
The sprite will not execute the last move.
The sprite will return to its starting position.
The program will stop with an error.

Yes, the answer is correct. Score: 1

Accepted answer isThe final movement will continue in the direction established by the earlier turn.

14.

Which Scratch blocks in the program only change the sprite's position and never its direction?

1 Point
the two move 5 blocks
all three move blocks
turn and point
move 10 and point

Yes, the answer is correct. Score: 1

Accepted answer isall three move blocks

15.

A student accidentally swaps the positions of the turn 15 degrees block and the point in direction 90° block. What is the most likely effect?

1 Point
The sprite will follow exactly the same path.
The final movement will occur after turning 15° from 90°, producing a different path.
The sprite will not execute the second movement.
The program will become an infinite loop.

Yes, the answer is correct. Score: 1

Accepted answer isThe final movement will occur after turning 15° from 90°, producing a different path.

Case Study: Evaluating a Scratch-Based Game Scoring System

A student is developing a simple game in Scratch to reward players based on their performance. The Scratch program shown in the figure keeps track of three variables: score, counter, and bonus.

When the green flag is clicked, the variables are initialized. The program then repeats a set of instructions several times. During each repetition, the player's score increases, the round counter is updated, and a bonus is awarded depending on whether the current round number is even or odd. After all repetitions are complete, the accumulated bonus is added to the score.

Finally, the program displays one of three messages—"Excellent!", "Level Cleared!", or

"Try Again!"—depending on the final score. Study the Scratch program carefully and answer the following questions.



16.

Why does the program use the expression counter mod 2 = 0?

1 Point

To check whether the current round number is even.
To stop the repeat loop after two iterations.
To calculate the total score.
To reset the bonus variable.

Yes, the answer is correct. Score: 1

Accepted answer isTo check whether the current round number is even.

17.

Which variable is updated in every iteration of the repeat loop, regardless of the outcome of the if-else condition?

1 Point

bonus
counter
score
Both score and counter

Yes, the answer is correct. Score: 1

Accepted answer isBoth score and counter

18.

What is the purpose of the statement change score by bonus placed after the repeat loop?

1 Point

It resets the score before displaying the final message.
It adds the total bonus accumulated during all iterations to the final score.
It increases the bonus by the current score.
It determines how many times the loop executes.

Yes, the answer is correct. Score: 1

Accepted answer isIt adds the total bonus accumulated during all iterations to the final score.

19.

Suppose the statement change score by bonus is moved inside the repeat loop, immediately before the if condition. What would be the most likely effect?

1 Point

The final score would become much larger because the bonus would be added repeatedly.
The final score would remain unchanged.
The bonus variable would always become zero.
The repeat loop would execute fewer times.

Yes, the answer is correct. Score: 1

Accepted answer isThe final score would become much larger because the bonus would be added repeatedly.

20.

Which statement best describes how the Scratch program decides which message to display?

1 Point

The message depends only on the value of counter.
The message depends only on the value of bonus.
The program first computes the final score and then checks the score against the conditions in order.
All three messages are displayed one after another.

Yes, the answer is correct. Score: 1

Accepted answer isThe program first computes the final score and then checks the score against the conditions in order.

Week 2 : Assignment 2

Due date on 2026-08-05, 23:59 IST

The due date for submitting this assignment has passed.

Your last recorded submission was on 2026-07-28, 22:17 IST.

Scenario: Student Registration for a Coding Bootcamp

A school is organizing a one-day coding bootcamp for beginners. When students arrive, they register at a computer kiosk. The kiosk asks each student to enter their name and age. It stores this information so it can display a personalized welcome message and generate a participation slip. During registration, if a student notices that they entered incorrect information, they can update it before the final slip is printed. The computer always uses the most recently entered information when displaying the student's details.

1.

Why does the registration kiosk ask each student to enter their name?

1 Point

To personalize the welcome message and participation slip.
To make the computer process information faster.
To install the coding software.
To increase the computer's storage.

Yes, the answer is correct. Score: 1

Accepted answer isTo personalize the welcome message and participation slip.

2.

A student first enters their age as 15, but later realizes it should be 16 and updates it before the participation slip is printed. Which age will appear on the slip?

1 Point

15
Both 15 and 16
16
No age will be displayed.

Yes, the answer is correct. Score: 1

Accepted answer is16

3.

The computer displays a greeting that includes the student's name without asking for it again. How is this possible?

1 Point

The computer remembers every student's name forever.
The name is stored and reused whenever needed during the program.
The greeting works only for the first student.
The student secretly enters the name again.

Yes, the answer is correct. Score: 1

Accepted answer isThe name is stored and reused whenever needed during the program.

4.

A student accidentally enters the name Riya instead of Diya. Before the participation slip is printed, the mistake is corrected. What is the most likely outcome?

1 Point

The slip will show Riya.
The slip will show both Riya and Diya.
The slip will show Diya.
The slip will not display any name.

Yes, the answer is correct. Score: 1

Accepted answer isThe slip will show Diya.

5.

Why is it useful for the program to store information like a student's name and age?

1 Point

It allows the same information to be used multiple times without asking the student again.
It permanently stores the information on every computer.
It prevents the program from displaying messages.
It eliminates the need for user input in all future programs.

Yes, the answer is correct. Score: 1

Accepted answer isIt allows the same information to be used multiple times without asking the student again.

Scenario: Ordering Food at a School Cafeteria

A school cafeteria has introduced a self-service ordering kiosk. Students enter their name and the food item they would like to order. The kiosk stores this information and displays a confirmation message. Before the order is finalized, one student changes their order from **Sandwich** to **Pasta**. The kiosk updates the stored value and prints the final order receipt. The Python program running on the kiosk is shown below.

```
1. print("Welcome to the School Cafeteria!")
2. customer = input("Enter your name: ")
3. order = input("Enter your food item: ")
4. print("Hello", customer)
5. print("You ordered", order)
6. order = "Pasta"
7. print("Updated Order:", order)
8. print("Thank you for your order,", customer)
```

6.

Which line collects the name of the customer?

1 Point
Line 1
Line 2
Line 3
Line 8

Yes, the answer is correct. Score: 1

Accepted answer isLine 2

7.

A student initially enters Sandwich as the food item. What will be displayed in Line 7?

1 Point
Sandwich
Sandwich Pasta
Pasta
No food item will be displayed

Yes, the answer is correct. Score: 1

Accepted answer isPasta

8.

Which variable stores the name of the customer throughout the program?

1 Point
order
customer
input
print

Yes, the answer is correct. Score: 1

Accepted answer iscustomer

9.

If the customer enters Neha as the name, what will be displayed in Line 8?

1 Point
Thank you for your order, customer
Thank you for your order, Neha
Thank you for your order, Pasta
Thank you for your order

Yes, the answer is correct. Score: 1

Accepted answer isThank you for your order, Neha

10.

Why is the variable order updated in Line 6 instead of creating another variable?

1 Point

A variable can only store one value at a time, and the latest value replaces the previous one.
Python does not allow more than one variable in a program.
The original value cannot be changed.
Updating a variable permanently changes all future programs.

Yes, the answer is correct. Score: 1

Accepted answer isA variable can only store one value at a time, and the latest value replaces the previous one.

Scenario: Online Bookstore Checkout

An online bookstore is developing a simple checkout system. When a customer selects a book, the system asks for the book's price. Since the entered value comes from the keyboard, the system first converts it into a number before performing any calculations. A **10% discount** is then applied to the original price to calculate the final amount the customer has to pay.

While testing the program, the developers also discover that if they forget to convert the entered value into a number, mathematical calculations do not work as expected. They also learn that Python is very particular about following the correct syntax and indentation rules. Even a small mistake, such as forgetting a colon or incorrect indentation, can prevent the program from running.

11.

Why does the program convert the customer's entered price into a number before calculating the discount?

1 Point

To make the input appear on the screen.
To allow mathematical operations such as multiplication.
To permanently save the customer's purchase history.
To reduce the size of the program.

Yes, the answer is correct. Score: 1

Accepted answer isTo allow mathematical operations such as multiplication.

12.

A customer enters the price of a book as ₹800. After applying a 10% discount, what amount should the customer pay?

1 Point

₹720
₹780
₹880

₹700

Yes, the answer is correct. Score: 1

Accepted answer is₹720

13.

During testing, a developer forgets to convert the entered price into a number before performing calculations. What is the most likely outcome?

1 Point

The calculations may fail or produce unexpected results.
The discount becomes 20% automatically.
The program permanently deletes the input.
Python automatically fixes the mistake.

Yes, the answer is correct. Score: 1

Accepted answer isThe calculations may fail or produce unexpected results.

14.

Another developer accidentally removes the required colon (:) from a Python statement. What is the most likely result?

1 Point

The program will ignore the missing colon and continue.
Python will report a syntax error.
The program will apply the wrong discount.
The program will ask the user to re-enter the price.

Yes, the answer is correct. Score: 1

Accepted answer isPython will report a syntax error.

15.

Which of the following statements are correct about the bookstore's discount program?

1 Point

User input must often be converted into a number before performing arithmetic calculations.
Python requires correct syntax and indentation for a program to run properly.
A 10% discount means the customer pays 90% of the original price.
Python automatically treats every keyboard input as a number.

Yes, the answer is correct. Score: 1

Accepted answer isUser input must often be converted into a number before performing arithmetic calculations., Python requires correct syntax and indentation for a program to run properly., A 10% discount means the customer pays 90% of the original price.

Scenario: Morning Exercise Counter

A school has installed a digital exercise board in the playground. Every morning, the board displays the number of jumping jacks students should perform. It starts from 1 and continues until the teacher decides to stop the session. The board repeatedly displays the next exercise number as long as the teacher chooses to continue.

The following Python program is used for the exercise board.

```
1. print("Morning Exercise Begins")
2. count = 1
3. continue_session = 1
4.
5. while continue_session == 1:
6.     print("Do Exercise", count)
7.     count = count + 1
8.     continue_session = int(input("Continue? (1 = Yes, 0 = No): "
9.
10. print("Exercise Session Over")
```

16.

What is displayed first when the program starts?

1 Point

Do Exercise 1
Continue? (1 = Yes, 0 = No)
Morning Exercise Begins
Exercise Session Over

Yes, the answer is correct. Score: 1

Accepted answer isMorning Exercise Begins

17.

The teacher enters 1 three times and then enters 0. Which exercise numbers will be displayed?

1 Point

1, 2, 3
1, 2, 3, 4
0, 1, 2, 3
2, 3, 4

No, the answer is incorrect. Score: 0

Accepted answer is1, 2, 3, 4

18.

Which statements about the program are correct?

1 Point

The value of count increases by 1 during each repetition.
The while loop repeats only while continue_session is equal to 1.
The message "Exercise Session Over" is displayed after the loop ends.

The value of count decreases during each repetition.

Yes, the answer is correct. Score: 1

Accepted answer isThe value of count increases by 1 during each repetition., The while loop repeats only while continue session is equal to 1., The message "Exercise Session Over" is displayed after the loop ends.

19.

Why is a while loop used in this program?

1 Point

Because the number of repetitions depends on the teacher's choice.
Because the program always repeats exactly 10 times.
Because Python cannot print numbers without a while loop.
Because variables cannot be used inside a for loop.

Yes, the answer is correct. Score: 1

Accepted answer isBecause the number of repetitions depends on the teacher's choice.

20.

Which of the following statements correctly compare for loops and while loops?

1 Point

A for loop is useful when the number of repetitions is known in advance.
A while loop continues until its condition becomes false.
A while loop always executes exactly 10 times.
Both for and while loops are used to repeat tasks.

Yes, the answer is correct. Score: 1

Accepted answer isA for loop is useful when the number of repetitions is known in advance., A while loop continues until its condition becomes false., Both for and while loops are used to repeat tasks.

Week 3 : Assignment 3

Due date on 2026-08-14, 23:59 IST

The due date for submitting this assignment has passed.

Your last recorded submission was on 2026-08-01, 17:17 IST.

Scenario: Managing the School Library

A school library maintains a digital catalog of all its books using a Python list. As new books arrive, the librarian updates the catalog by adding new titles or inserting important reference books at specific positions. Sometimes multiple copies of the same book are available, so the librarian checks how many times a title appears in the list. To help students find books more easily, the catalog is sorted alphabetically. During special exhibitions, the sorted list is displayed in reverse alphabetical order. Occasionally, the librarian also displays only a selected range of books, such as books belonging to a particular shelf or category, instead of showing the entire catalog.

Answer the following questions based on the scenario.

1.

Why does the librarian sort the catalog of books?

1 Point

To remove duplicate book titles
To make the books easier to organize and locate
To increase the number of books in the catalog
To replace old books with new ones

Yes, the answer is correct. Score: 1

Accepted answer isTo make the books easier to organize and locate

2.

After sorting the catalog alphabetically, the librarian wants to display the books from Z to A. Which operation is most suitable?

1 Point

Count
Append
Reverse
Length

Yes, the answer is correct. Score: 1

Accepted answer isReverse

3.

The librarian wants to know how many copies of the book *Python Basics* are present in the catalog. Which list operation is most appropriate?

1 Point

Length
Reverse
Insert

Count

Yes, the answer is correct. Score: 1

Accepted answer isCount

4.

A newly purchased encyclopedia must appear as the second item in the catalog instead of being added at the end. Which operation should be used?

1 Point

Insert

Append

Sort

Reverse

Yes, the answer is correct. Score: 1

Accepted answer isInsert

5.

The librarian wants to know the total number of books currently stored in the catalog. Which operation provides this information?

1 Point

Count

Length

Reverse

Slice

Yes, the answer is correct. Score: 1

Accepted answer isLength

Scenario: Managing Participants for a Workshop

A training institute maintains a list of participant names for a Python workshop. As registrations are received, new participants are added to the list. A late VIP registration is inserted at a specific position. Before printing attendance sheets, the participant names are arranged alphabetically. Finally, only a selected group of participants is displayed for one workshop batch.

Study the following program and answer the questions.

```
1. participants = ["Riya", "Aman", "Kiran", "Neha", "Aman"]
2. participants.append("Sana")
3. participants.insert(1, "Bhavna")
4. participants.sort()
5. batch = participants[1:5]
6. print(participants)
7. print(batch)
```

6.

What will be the output of line 6?

1 Point

['Aman', 'Aman', 'Bhavna', 'Kiran', 'Neha', 'Riya', 'Sana']
['Riya', 'Bhavna', 'Aman', 'Kiran', 'Neha', 'Aman', 'Sana']
['Aman', 'Bhavna', 'Kiran', 'Neha', 'Riya', 'Sana']
['Bhavna', 'Aman', 'Aman', 'Kiran', 'Neha', 'Riya', 'Sana']

Yes, the answer is correct. Score: 1

Accepted answer is['Aman', 'Aman', 'Bhavna', 'Kiran', 'Neha', 'Riya', 'Sana']

7.

What is stored in the variable batch after line 5?

1 Point

['Aman', 'Bhavna', 'Kiran', 'Neha']
['Bhavna', 'Kiran', 'Neha', 'Riya']
['Aman', 'Aman', 'Bhavna', 'Kiran']
['Kiran', 'Neha', 'Riya', 'Sana']

Yes, the answer is correct. Score: 1

Accepted answer is['Aman', 'Bhavna', 'Kiran', 'Neha']

8.

Which of the following statements are true after line 4?

1 Point

The participant Sana has been added to the list.
The name Aman appears twice.
The list contains 6 participants.
The names are arranged alphabetically.

Yes, the answer is correct. Score: 1

Accepted answer isThe participant Sana has been added to the list., The name Aman appears twice., The names are arranged alphabetically.

9.

Which line inserts the participant Bhavna at a specific position in the list?

1 Point
Line 1
Line 2
Line 3
Line 5

Yes, the answer is correct. Score: 1

Accepted answer isLine 3

10.

Which operation is demonstrated by line 5?

1 Point
Appending a new participant
Sorting the participant list
Counting duplicate names
Extracting a portion of the participant list

Yes, the answer is correct. Score: 1

Accepted answer isExtracting a portion of the participant list

Scenario: Traffic Signal Announcement System

A city traffic department is developing a digital announcement system for a pedestrian crossing. The system displays numbers from 1 to 30. To make the display more informative, it follows these rules:

- If the number is divisible by 3, it displays "Walk" instead of the number.
- If the number is divisible by 5, it displays "Wait" instead of the number.
- If the number is divisible by both 3 and 5, it displays "WalkWait".
- Otherwise, it displays the number itself.

A programmer uses a loop and conditional statements to implement this logic.
Answer the following questions based on the scenario.

11.

What should the system display when the number 15 is reached?

1 Point
15
Walk
Wait
WalkWait

Yes, the answer is correct. Score: 1

Accepted answer isWalkWait

12.

Which of the following numbers will display "Walk"?

1 Point

8
12
20
25

Yes, the answer is correct. Score: 1

Accepted answer is12

13.

A programmer first checks whether a number is divisible by 3, then checks whether it is divisible by 5, and only after that checks whether it is divisible by both 3 and 5. What problem is likely to occur?

1 Point

Numbers divisible by both 3 and 5 will be displayed as Walk instead of WalkWait.
Numbers divisible by 5 will never be displayed.
The loop will never stop.
The program will print every number twice.

Yes, the answer is correct. Score: 1

Accepted answer isNumbers divisible by both 3 and 5 will be displayed as Walk instead of WalkWait.

14.

Which programming concept allows the system to process every number from 1 to 30 automatically?

1 Point

Function
List
Loop
Variable

Yes, the answer is correct. Score: 1

Accepted answer isLoop

15.

Which statement best explains why the condition "divisible by both 3 and 5" should be checked before checking divisibility by 3 or 5 individually?

1 Point

It reduces the number of iterations in the loop.
It ensures that numbers satisfying both conditions receive the correct output.
It changes the range of numbers processed.
It makes the program run without using loops.

Yes, the answer is correct. Score: 1

Accepted answer isIt ensures that numbers satisfying both conditions receive the correct output.

Scenario: Choosing the Best Estimate for a Wildlife Survey

A wildlife research team wants to estimate the number of birds nesting in a protected forest. Since counting every bird is difficult, they ask 120 volunteers to independently estimate the number of nests. The estimates vary widely because some volunteers underestimate while

others overestimate the actual value. The researchers analyze the estimates by sorting them, removing extreme values, and computing different aggregate measures. They also visualize the estimates using Matplotlib to better understand the spread of the data.

Answer the following questions based on the scenario.

16.

Why did the researchers ask a large number of volunteers instead of only one wildlife expert?

1 Point

To reduce the size of the dataset
Because a larger group can produce a reliable collective estimate
Because volunteers always know the exact answer
To avoid using statistical measures

Yes, the answer is correct. Score: 1

Accepted answer isBecause a larger group can produce a reliable collective estimate

17.

Which statistical measure is specifically designed to reduce the effect of extremely high and extremely low estimates?

1 Point

Maximum
Median
10% Trimmed Mean
Range

Yes, the answer is correct. Score: 1

Accepted answer is10% Trimmed Mean

18.

The researchers create a histogram of all the estimates using Matplotlib. What is the main purpose of this graph?

1 Point

To sort the estimates automatically
To visualize how the estimates are distributed
To calculate the trimmed mean
To remove incorrect estimates

Yes, the answer is correct. Score: 1

Accepted answer isTo visualize how the estimates are distributed

19.

Which of the following statements about 10% trimmed mean are correct?

1 Point

The data should be sorted before removing extreme values.
The smallest and largest 10% of estimates are removed.
Every estimate is included in the final calculation.
It helps reduce the influence of outliers.

Yes, the answer is correct. Score: 1

Accepted answer isThe data should be sorted before removing extreme values., The smallest and largest 10% of estimates are removed., It helps reduce the influence of outliers.

20.

The researchers use the following statement before calculating the trimmed mean:

estimates.sort()

Which of the following are true about this statement?

1 Point

It arranges the estimates in ascending order.
It changes the original list of estimates.
It creates a new sorted copy while leaving the original unchanged.
It helps identify the smallest and largest estimates for trimming.

Yes, the answer is correct. Score: 1

Accepted answer isIt arranges the estimates in ascending order., It changes the original list of estimates., It helps identify the smallest and largest estimates for trimming.

Week 4 : Assignment 4

Due date on 2026-08-19, 23:59 IST

The due date for submitting this assignment has passed.

Your last recorded submission was on 2026-08-10, 19:26 IST.

Scenario: Birthday Collision Simulator

A data science club is developing a Python program to demonstrate the Birthday Paradox, which shows that in a surprisingly small group of people, two individuals are likely to share the same birthday. Since collecting real birthday data is impractical, the program randomly generates birthdays for 50 people. It first generates a random year, month, and day while ensuring that the generated date is valid by checking whether the selected year is a leap year. The birthdays are stored as datetime.date objects and then converted into their corresponding day of the year (1–365 or 366), since only the day and month are relevant for detecting birthday matches. These values are stored in a list, sorted in ascending order, and scanned for repeated values. Each repeated value represents a birthday collision. Because the birthdays are generated randomly, running the program multiple times produces different numbers of collisions, helping students understand the Birthday Paradox through simulation.

1.

Why does the program check whether a randomly generated year is a leap year?

1 Point

To ensure that February is assigned the correct number of days.
To reduce the number of duplicate birthdays.
To generate birthdays in chronological order.
To calculate the person's age.

Yes, the answer is correct. Score: 1

Accepted answer isTo ensure that February is assigned the correct number of days.

2.

Which Python library is primarily responsible for creating valid calendar dates in the program?

1 Point

random
math
datetime
calendar

Yes, the answer is correct. Score: 1

Accepted answer isdatetime

3.

Why does the program convert each birthday into its "day of the year" value?

1 Point

To calculate the age of each person.
To compare birthdays without considering the birth year.

To reduce the amount of memory used.
To make sorting unnecessary.

Yes, the answer is correct. Score: 1

Accepted answer isTo compare birthdays without considering the birth year.

4.

Why is the list of day-of-year values sorted before checking for collisions?

1 Point

To generate birthdays more efficiently.
To remove invalid dates automatically.
To arrange birthdays by year.
To make duplicate values easier to identify.

Yes, the answer is correct. Score: 1

Accepted answer isTo make duplicate values easier to identify.

5.

Why does the program usually produce a different number of collisions each time it is executed?

1 Point

The random birthday generation creates a different set of birthdays in every run.
The leap year calculation changes during execution.
The sorting algorithm produces a different order every time.
The datetime library stores dates differently in each execution.

Yes, the answer is correct. Score: 1

Accepted answer isThe random birthday generation creates a different set of birthdays in every run.

Scenario: Digital Dobble Card Generator

A game development team is creating a digital version of the Dobble card game using Python. Each card contains five symbols, represented by letters of the alphabet. The program randomly generates two cards such that exactly one symbol is common between them, while all other symbols are unique. To achieve this, it uses Python's string library to obtain a pool of symbols and the random library to select symbols and their positions. Once a symbol is placed on a card, it is removed from the available pool to prevent duplicates. After generating the cards, the program displays them and asks the player to identify the common symbol. If the player's input matches the common symbol, the program prints "Correct"; otherwise, it prints "Wrong". During testing, the developers also discovered that generating random indices from 0 to 5 for a five-element list caused an IndexError, which was fixed by generating indices from 0 to 4.

6.

What is the primary objective of the Python program?

1 Point

To generate two cards with exactly one common symbol.
To sort symbols alphabetically before displaying them.

To count the number of vowels on each card.
To generate identical cards every time.

Yes, the answer is correct. Score: 1

Accepted answer isTo generate two cards with exactly one common symbol.

7.

Which of the following steps help ensure that the two cards contain exactly one common symbol?

1 Point

Removing the common symbol from the symbol pool after placing it.
Removing every newly assigned symbol from the available symbol pool.
Randomly choosing only one symbol to be shared between both cards.
Sorting the cards after they are generated.

Yes, the answer is correct. Score: 1

Accepted answer isRemoving the common symbol from the symbol pool after placing it., Removing every newly assigned symbol from the available symbol pool., Randomly choosing only one symbol to be shared between both cards.

8.

Why does the program use random.choice()?

1 Point

To sort the available symbols.
To select a symbol randomly from the available symbol list.
To compare the two cards.
To convert uppercase letters into lowercase letters.

Yes, the answer is correct. Score: 1

Accepted answer isTo select a symbol randomly from the available symbol list.

9.

Why was random.randint(0, 5) changed to random.randint(0, 4) while selecting card positions?

1 Point

The cards contain only four symbols.
The random library cannot generate the value 5.
A five-element list has valid indices only from 0 to 4.
Python lists always begin with index 1.

Yes, the answer is correct. Score: 1

Accepted answer isA five-element list has valid indices only from 0 to 4.

10.

What does the program do after displaying both generated cards?

1 Point

It automatically highlights the common symbol.
It asks the player to identify the common symbol.
It generates another pair of cards immediately.
It removes all symbols from both cards.

Yes, the answer is correct. Score: 1

Accepted answer isIt asks the player to identify the common symbol.

Scenario: Building the First Prototype of a Magic Square Generator

A software development team is creating a Python program to generate odd-sized Magic Squares. Rather than implementing the complete algorithm immediately, they decide to build and test the program incrementally. In the first prototype, the program accepts the size of the square (n), creates an n × n matrix initialized with zeros, determines the starting position for the first number using the Magic Square algorithm, places the number 1 at that position, and then displays the matrix for verification. Once this initial step works correctly, the team plans to extend the program to place the remaining numbers by following the movement rules of the algorithm.

```
n = 3

magic_square = [[0 for j in range(n)] for i in range(n)]

row = n // 2
col = n - 1

magic_square[row][col] = 1

for r in magic_square:
    print(r)
```

11.

Which programming construct is primarily responsible for creating the two-dimensional matrix in the prototype?

1 Point

Nested list comprehension
Nested if-else statements
Nested while loops
Function recursion

Yes, the answer is correct. Score: 1

Accepted answer isNested list comprehension

12.

If the value of n is changed from 3 to 5, which statement remains true without modifying the rest of the program?

1 Point

The number 1 will always be placed in the center of the matrix.
The number 1 will always be placed in the middle row of the last column.
The matrix will automatically become a valid Magic Square.
The program will generate numbers from 1 to 25 automatically.

Yes, the answer is correct. Score: 1

Accepted answer isThe number 1 will always be placed in the middle row of the last column.

13.

Why is the matrix initialized with zeros instead of leaving it empty?

1 Point

To allow the program to identify cells that have not yet been assigned a value.
To ensure that every row initially has the same sum.
To improve the speed of printing the matrix.
To prevent duplicate numbers from being generated.

Yes, the answer is correct. Score: 1

Accepted answer isTo allow the program to identify cells that have not yet been assigned a value.

14.

What is the primary limitation of this prototype compared to the complete Magic Square algorithm?

1 Point

It cannot create matrices larger than 3 × 3.
It places only the first number and does not determine positions for the remaining numbers.
It cannot display the matrix.
It requires recursion to execute.

Yes, the answer is correct. Score: 1

Accepted answer isIt places only the first number and does not determine positions for the remaining numbers.

15.

Which of the following statements correctly describe the prototype program?

1 Point

It allocates memory for an n × n matrix before placing any numbers.
It determines the initial position of the first number using the Magic Square algorithm.
It uses nested iteration while creating the matrix.
It verifies that every row, column, and diagonal has the same sum.

Yes, the answer is correct. Score: 1

Accepted answer isIt allocates memory for an n × n matrix before placing any numbers., It determines the initial position of the first number using the Magic Square algorithm., It uses nested iteration while creating the matrix.

Scenario: Movie Guessing Game

A movie streaming platform is developing a Movie Guessing Game to improve user engagement. The game randomly selects a movie title from a predefined list and displays a masked version of it, where every alphabetic character is replaced with * while spaces remain visible. The player repeatedly guesses one letter at a time. If the guessed letter is present in the movie title, all its occurrences are revealed. Otherwise, the player is informed that the letter is not present. The game continues until the player successfully reveals the complete movie title.

```
import random

movies = [
    "THE LION KING",
    "3 IDIOTS",
    "DANGAL",
    "LAGAAN",
    "INTERSTELLAR"
]


def create_question(movie):
    question = ""

    for ch in movie:
        if ch == " ":
            question += " "
        elif ch.isalpha():
            question += "*"
        else:
            question += ch

    return question
```

```
def is_present(letter, movie):
    return movie.count(letter) > 0


def unlock(question, movie, letter):
    result = ""

    for i in range(len(movie)):
        if movie[i] == letter:
            result += letter
        else:
            result += question[i]

    return result


movie = random.choice(movies)

question = create_question(movie)
```

```
print("Guess the movie:")
print(question)

while question != movie:

    letter = input("Enter a letter: ").upper()

    if is_present(letter, movie):
        question = unlock(question, movie, letter)
        print("Correct!")
    else:
        print("Letter not found.")

    print(question)

print("Congratulations! You guessed the movie.")
```

16.

Which function is responsible for creating the initial masked movie title displayed to the player?

1 Point

unlock()
is_present()
create_question()
random.choice()

Yes, the answer is correct. Score: 1

Accepted answer iscreate_question()

17.

The player enters a letter that is not present in the selected movie title. What happens next?

1 Point

The movie title is selected again.
The current masked title remains unchanged, and a message is displayed.
The game terminates immediately.
All hidden letters are revealed.

Yes, the answer is correct. Score: 1

Accepted answer isThe current masked title remains unchanged, and a message is displayed.

18.

Why does the unlock() function receive both the original movie title and the current masked title as parameters?

1 Point

To compare their lengths before displaying the result.
To generate a new random movie title.
To reveal only the newly guessed letters while preserving previously revealed letters.
To convert the movie title into uppercase.

Yes, the answer is correct. Score: 1

Accepted answer isTo reveal only the newly guessed letters while preserving previously revealed letters.

19.

Which condition causes the game loop to terminate?

1 Point

The player enters an incorrect letter.
All letters in the movie title have been revealed.
The player guesses five incorrect letters.
The movie list becomes empty.

Yes, the answer is correct. Score: 1

Accepted answer isAll letters in the movie title have been revealed.

20.

Which of the following statements about the program are correct?

1 Point

random.choice() is used to select one movie title from the list.
isalpha() ensures that spaces and non-alphabetic characters are replaced with *.
The unlock() function may reveal multiple occurrences of the guessed letter in a single attempt.
The player's input is converted to uppercase before it is checked against the movie title.

Yes, the answer is correct. Score: 1

Accepted answer israndom.choice() is used to select one movie title from the list., The unlock() function may reveal multiple occurrences of the guessed letter in a single attempt., The player's input is converted to uppercase before it is checked against the movie title.

Week 5 : Assignment 5

Due date on 2026-08-26, 23:59 IST

The due date for submitting this assignment has passed.

Your last recorded submission was on 2026-08-21, 21:56 IST.

Scenario: Community Library Book Tracker

A community library keeps track of the number of books borrowed from different categories. At the end of the day, the librarian wants to update the records based on newly borrowed books and determine which category had the highest number of borrowings. Study the following Python program carefully and answer the questions that follow.

```
1.  borrow_count = {
2.      "Fiction": 12,
3.      "Science": 8,
4.      "History": 5,
5.      "Technology": 10
6.  }

7.  updates = {
8.      "Science": 4,
9.      "History": 2,
10.     "Art": 6
11. }

12. for category, count in updates.items():
13.     if category in borrow_count:
14.         borrow_count[category] += count
15.     else:
16.         borrow_count[category] = count

17. highest = ""
18. max_books = -1

19. for category, count in borrow_count.items():
20.     if count > max_books:
21.         max_books = count
22.         highest = category

23. print(borrow_count)
24. print(highest, max_books)
```

1.

How many key-value pairs are present in borrow_count immediately after the completion of Line 16?

1 Point

4

5

6
7

Yes, the answer is correct. Score: 1

Accepted answer is5

2.

Which statement correctly describes the execution of the else block (Lines 15–16)?

1 Point

It executes for every category in updates.
It executes only when a category already exists in borrow_count.
It executes only when a category is not present in borrow_count.
It never executes because all categories already exist.

Yes, the answer is correct. Score: 1

Accepted answer isIt executes only when a category is not present in borrow_count.

3.

Immediately after executing Line 21 for the first time, what is the value of max_books?

1 Point

-1
10
12
7

Yes, the answer is correct. Score: 1

Accepted answer is12

4.

Which variable stores the largest borrowing count after the program completes execution?

1 Point

highest
category
count
max_books

Yes, the answer is correct. Score: 1

Accepted answer ismax_books

5.

Which of the following statements are correct?

1 Point

The values of "Science" and "History" are increased.
The value of "Technology" remains unchanged.
Exactly one new key is added to the dictionary.
The final dictionary contains fewer keys than the initial dictionary.

Yes, the answer is correct. Score: 1

Accepted answer isThe values of "Science" and "History" are increased., The value of "Technology" remains unchanged., Exactly one new key is added to the dictionary.

Scenario: Rock–Paper–Scissors Championship

A coding club is organizing a Rock–Paper–Scissors championship. The choices made by both players are stored in two lists. A dictionary is used to define which move defeats another. The program compares the choices in each round, updates the scores, and declares the overall winner.

Study the following program and answer the questions.

```
wins_against = {
    "Rock": "Scissors",
    "Paper": "Rock",
    "Scissors": "Paper"
}

playerA = ["Rock", "Paper", "Scissors", "Rock", "Paper"]
playerB = ["Scissors", "Rock", "Paper", "Rock", "Scissors"]

score = {
    "Player A": 0,
    "Player B": 0
}

for i in range(len(playerA)):
    if playerA[i] == playerB[i]:
        continue
    elif wins_against[playerA[i]] == playerB[i]:
        score["Player A"] += 1
    else:
        score["Player B"] += 1

print(score)

if score["Player A"] > score["Player B"]:
    print("Player A Wins")
elif score["Player B"] > score["Player A"]:
    print("Player B Wins")
else:
    print("Match Draw")
```

6.

What is the purpose of the dictionary wins_against?

1 Point

It stores the scores of both players.
It stores the choices made by each player.
It stores which move defeats another move.
It stores the winner of each round.

Yes, the answer is correct. Score: 1

Accepted answer isIt stores which move defeats another move.

7.

What is the value of the score dictionary after the loop finishes?

1 Point
{"Player A": 2, "Player B": 2}
{"Player A": 3, "Player B": 1}
{"Player A": 4, "Player B": 0}
{"Player A": 1, "Player B": 3}

Yes, the answer is correct. Score: 1

Accepted answer is{"Player A": 3, "Player B": 1}

8.

What does the expression wins_against[playerA[i]] return?

1 Point
The current score of Player A.
The move that defeats Player A's current move.
The move defeated by Player A's current move.
The current round number.

Yes, the answer is correct. Score: 1

Accepted answer isThe move defeated by Player A's current move.

9.

Which statement is printed at the end of the program?

1 Point
Player A Wins
Player B Wins
Match Draw
No output is printed

Yes, the answer is correct. Score: 1

Accepted answer isPlayer A Wins

10.

Which of the following statements are correct?

1 Point
The dictionary score is updated during program execution.
The dictionary wins_against is modified during the program.
The program records one drawn round.
The statement score["Player A"] += 1 updates the value associated with the key "Player A".

Yes, the answer is correct. Score: 1

Accepted answer isThe dictionary score is updated during program execution., The program records one drawn round., The statement score["Player A"] += 1 updates the value associated with the key "Player A".

Library Book Search System

A digital library stores book IDs in a sorted list. When a student searches for a book, the librarian first performs a Linear Search to locate the book. Later, the system uses a Binary Search to improve search efficiency on the same sorted list.

Study the following program and answer the questions.

```
1. book_ids = [105, 112, 118, 126, 134, 142, 150, 163]
2. target = 134
3.
4. # Linear Search
5. linear_index = -1
6. for i in range(len(book_ids)):
7.     if book_ids[i] == target:
8.         linear_index = i
9.         break
10.
11. # Binary Search
12. low = 0
13. high = len(book_ids) - 1
14. binary_index = -1
15.
16. while low <= high:
17.     mid = (low + high) // 2
18.
19.     if book_ids[mid] == target:
20.         binary_index = mid
21.         break
22.     elif target < book_ids[mid]:
23.         high = mid - 1
24.     else:
25.         low = mid + 1
26.
27. print("Linear Search:", linear_index)
28. print("Binary Search:", binary_index)
```

11.

Why is the list book_ids arranged in ascending order?

1 Point

To reduce memory usage.
Because Binary Search requires the data to be sorted.
Because Linear Search requires the data to be sorted.
To avoid duplicate values.

Yes, the answer is correct. Score: 1

Accepted answer isBecause Binary Search requires the data to be sorted.

12.

What is the output of the program?

1 Point

Linear Search: 3 and Binary Search: 3
Linear Search: 4 and Binary Search: 4
Linear Search: 5 and Binary Search: 5
Linear Search: -1 and Binary Search: -1

Yes, the answer is correct. Score: 1

Accepted answer isLinear Search: 4 and Binary Search: 4

13.

During the first iteration of the Binary Search loop, what is the value of mid?

1 Point
2
3
4
5

Yes, the answer is correct. Score: 1

Accepted answer is3

14.

Suppose the statement break on Line 9 is removed. Which statement is correct?

1 Point
The program produces an error.
The Linear Search continues checking the remaining elements even after finding the target.
The Binary Search stops working.
The value of linear_index becomes -1.

Yes, the answer is correct. Score: 1

Accepted answer isThe Linear Search continues checking the remaining elements even after finding the target.

15.

Which of the following statements are correct?

1 Point
Linear Search checks elements one by one from the beginning of the list.
Binary Search repeatedly divides the search space into two halves.
Both Linear Search and Binary Search return index 4 for the given input.
Binary Search can be applied correctly even if book_ids is not sorted.

Yes, the answer is correct. Score: 1

Accepted answer isLinear Search checks elements one by one from the beginning of the list., Binary Search repeatedly divides the search space into two halves., Both Linear Search and Binary Search return index 4 for the given input.

Scenario: Airport Boarding Gate Search

An airport maintains a sorted list of boarding gate numbers for departing flights.

[101, 104, 108, 112, 117, 121, 126, 130, 135, 140]

A passenger is looking for Gate 126. The airport system can use either Linear Search or Binary Search to locate the gate.

Answer the following questions.

16.

How many elements would Linear Search examine before finding Gate 126?

1 Point
5
6
7
8

No, the answer is incorrect. Score: 0

Accepted answer is7

17.

During the first comparison of Binary Search, which gate number is examined?

1 Point

112
117
121
126

Yes, the answer is correct. Score: 1

Accepted answer is117

18.

After the first Binary Search comparison, which part of the list will be searched next?

1 Point

[101, 104, 108, 112]
[101, 104, 108, 112, 117]
[121, 126, 130, 135, 140]
[126, 130, 135, 140]

Yes, the answer is correct. Score: 1

Accepted answer is[121, 126, 130, 135, 140]

19.

Why is Binary Search generally faster than Linear Search for large sorted lists?

1 Point

It compares every element only once.
It checks two elements in every comparison.
It eliminates half of the remaining search space after each comparison.
It works only on lists with unique values.

Yes, the answer is correct. Score: 1

Accepted answer isIt eliminates half of the remaining search space after each comparison.

20.

Which of the following statements are correct?

1 Point

Binary Search requires the list to be sorted.
Linear Search can be used on both sorted and unsorted lists.
Both Linear Search and Binary Search will locate Gate 126 in this scenario.
Binary Search always examines every element in the list.

Yes, the answer is correct. Score: 1

Accepted answer isBinary Search requires the list to be sorted., Linear Search can be used on both sorted and unsorted lists., Both Linear Search and Binary Search will locate Gate 126 in this scenario.

Week 6 : Assignment 6

Due date on 2026-09-02, 23:59 IST

The due date for submitting this assignment has passed.

Your last recorded submission was on 2026-08-21, 22:01 IST.

Scenario: Secret Notice Encoder

A college club regularly shares announcements with its members. Since these messages are posted on a public notice board, the club wants to make them difficult for outsiders to read. To achieve this, the club uses a Substitution Cipher. Every alphabetic character is replaced by the previous letter in string.ascii_letters. Non-alphabetic characters such as spaces, digits, and punctuation marks remain unchanged.

The following Python program performs this encryption.

```
1. import string
2.
3. cipher = {}
4.
5. letters = string.ascii_letters
6.
7. for i in range(len(letters)):
8.     cipher[letters[i]] = letters[i - 1]
9.
10. text = input("Enter a message: ")
11.
12. encrypted = ""
13.
14. for ch in text:
15.     if ch in cipher:
16.         encrypted += cipher[ch]
17.     else:
18.         encrypted += ch
19.
20. print("Original :", text)
21. print("Encrypted:", encrypted)
```

1.

Which line creates the substitution rule that maps each letter to its replacement?

1 Point

Line 5

Line 7

Line 8
Line 16

Yes, the answer is correct. Score: 1

Accepted answer isLine 8

2.

Suppose the user enters the input:

ABC123

Which statement ensures that the digits 1, 2, and 3 remain unchanged?

1 Point
Line 10
Lines 17–18
Line 20
Line 21

Yes, the answer is correct. Score: 1

Accepted answer isLines 17–18

3.

What is the purpose of Line 12?

1 Point
It stores the substitution dictionary.
It stores the original input.
It initializes an empty string to build the encrypted message.
It prints the encrypted output.

Yes, the answer is correct. Score: 1

Accepted answer isIt initializes an empty string to build the encrypted message.

4.

If Line 16

encrypted += cipher[ch]

is replaced with

encrypted += ch

what will be the result?

1 Point
Only uppercase letters will be encrypted.
The program will generate an error.
Only digits will be encrypted.
The output will be identical to the input.

Yes, the answer is correct. Score: 1

Accepted answer isThe output will be identical to the input.

5.

Which of the following statements about the program are correct?

1 Point
Line 5 stores all lowercase and uppercase English letters.
Line 8 creates a substitution mapping using a dictionary.
Line 15 checks whether the current character is a digit.

Lines 20 and 21 display both the original and encrypted messages.

Yes, the answer is correct. Score: 1

Accepted answer isLine 5 stores all lowercase and uppercase English letters., Line 8 creates a substitution mapping using a dictionary., Lines 20 and 21 display both the original and encrypted messages.

Scenario: Tic-Tac-Toe Winner Checker

A programming club is developing a simple Tic-Tac-Toe game. The game board is represented using a nested list, where each inner list represents a row of the board.

- 'X' represents Player X.
- 'O' represents Player O.
- ' ' (space) represents an empty cell.

The program checks only the rows of the board to determine whether either player has won. If a row contains three identical non-space symbols, that player is declared the winner. Otherwise, the program reports that there is no row winner.

```
1. board = [
2.     ['X', 'X', 'X'],
3.     ['O', ' ', 'O'],
4.     [' ', 'O', ' ']
5. ]
6.
7. winner = "None"
8.
9. for row in board:
10.     if row[0] == row[1] == row[2] and row[0] != ' ':
11.         winner = row[0]
12.         break
13.
14. if winner != "None":
15.     print("Winner:", winner)
16. else:
17.     print("No row winner")
```

6.

Which line checks whether all three cells in a row contain the same non-empty symbol?

1 Point
Line 7
Line 9
Line 10
Line 15

Yes, the answer is correct. Score: 1

Accepted answer isLine 10

7.

What is the purpose of the break statement in Line 12?

1 Point
It skips the current row.
It exits the loop after finding a winner.
It resets the board.
It starts checking the rows again.

Yes, the answer is correct. Score: 1

Accepted answer isIt exits the loop after finding a winner.

8.

Which variable stores the winning player's symbol?

1 Point
board
row
winner
print

Yes, the answer is correct. Score: 1

Accepted answer iswinner

9.

If Line 11

winner = row[0]

is replaced with

winner = "Player"

what will the program display when a winning row is found?

1 Point
The winning symbol (X or O)
"Player"
"None"
The entire row

Yes, the answer is correct. Score: 1

Accepted answer is"Player"

10.

Which of the following statements about the program are correct?

1 Point
The board is represented using a nested list.
The program checks only row-wise winning conditions.

The loop stops immediately after a winner is found.
The program checks both row-wise and diagonal winning conditions.

Yes, the answer is correct. Score: 1

Accepted answer isThe board is represented using a nested list., The program checks only row-wise winning conditions., The loop stops immediately after a winner is found.

Scenario-Based Question: Recursive Folder Search

A company stores its documents in a hierarchy of folders. Each folder can contain:

- Files
- Other folders

A software engineer is asked to write a program to find a file named Report.pdf. The program begins with the main folder. Whenever it encounters a subfolder, it performs the same search process inside that subfolder. If the file is found, the search stops. If a folder contains no more subfolders or files, the program returns to the previous folder and continues searching. No loops are explicitly used to revisit folders; instead, the same search procedure is repeatedly applied to each subfolder.

11.

Which characteristic of the search process best represents recursion?

1 Point

The search visits folders in alphabetical order.
The same search procedure is applied repeatedly to each subfolder.
Every folder contains the same number of files.
The search uses multiple computers simultaneously.

Yes, the answer is correct. Score: 1

Accepted answer isThe same search procedure is applied repeatedly to each subfolder.

12.

In this scenario, what is the base case for the recursive search?

1 Point

The search begins from the main folder.
A folder contains only PDF files.
A folder has no more files or subfolders to search, or the required file is found.
The search enters a subfolder.

Yes, the answer is correct. Score: 1

Accepted answer isA folder has no more files or subfolders to search, or the required file is found.

13.

Why is recursion a suitable approach for this problem?

1 Point

Every folder has exactly one subfolder.
The same search operation must be repeated for each subfolder.
Recursion is always faster than iteration.

Files can only exist in the main folder.

Yes, the answer is correct. Score: 1

Accepted answer isThe same search operation must be repeated for each subfolder.

14.

What would most likely happen if the search process had no base case?

1 Point

The search would immediately stop.
The search would examine only the main folder.
The search could continue indefinitely or until system resources are exhausted.
The search would automatically switch to an iterative approach.

Yes, the answer is correct. Score: 1

Accepted answer isThe search could continue indefinitely or until system resources are exhausted.

15.

Which of the following statements correctly describe recursion in the given scenario?

1 Point

The same search procedure is used for every subfolder.
The search must have a stopping condition.
Each subfolder represents a smaller instance of the original problem.
Recursion can only be used if every folder contains exactly one subfolder.

Yes, the answer is correct. Score: 1

Accepted answer isThe same search procedure is used for every subfolder., The search must have a stopping condition., Each subfolder represents a smaller instance of the original problem.

Scenario: Recursive Binary Search for Student Roll Numbers

A university stores student roll numbers in ascending order. When an administrator wants to verify whether a particular roll number exists, the system performs a recursive binary search.
The search begins by checking the middle roll number in the current range.

- If the middle roll number matches the required roll number, the search is successful.
- If the required roll number is smaller, the same search process is repeated on the left half.
- If the required roll number is larger, the search continues recursively on the right half.
- If the search range becomes invalid (the left index becomes greater than the right index), the roll number is reported as not found.

The following program implements this recursive binary search.

```
1. def binary_search(arr, low, high, target):
2.     if low > high:
3.         return -1
4.
5.     mid = (low + high) // 2
6.
7.     if arr[mid] == target:
8.         return mid
9.
10.    elif target < arr[mid]:
11.        return binary_search(arr, low, mid - 1, target)
12.
13.    else:
14.        return binary_search(arr, mid + 1, high, target)
15.
16. numbers = [105, 112, 118, 124, 131, 145, 152, 168]
17. key = 131
18.
19. result = binary_search(numbers, 0, len(numbers)-1, key)
20.
21. if result != -1:
22.     print("Roll number found at index", result)
23. else:
24.     print("Roll number not found")
```

16.

Which line represents the base case where the search terminates because the search range has become invalid?

1 Point

Line 5

Line 2

Line 7

Line 14

Yes, the answer is correct. Score: 1

Accepted answer isLine 2

17.

Which line performs the recursive search on the left half of the list?

1 Point
Line 5
Line 8
Line 11
Line 19

Yes, the answer is correct. Score: 1

Accepted answer isLine 11

18.

What is the purpose of Line 5?

1 Point
It checks whether the element is found.
It calculates the middle index of the current search range.
It sorts the list.
It prints the search result.

Yes, the answer is correct. Score: 1

Accepted answer isIt calculates the middle index of the current search range.

19.

If Line 14

return binary_search(arr, mid + 1, high, target)

is replaced with

return binary_search(arr, low, high, target)

what is the most likely outcome when the target is greater than the middle element?

1 Point
The program always finds the target immediately.
The search correctly moves to the right half.
The search continues without reducing the search range, leading to infinite recursion.
The program searches only the left half.

Yes, the answer is correct. Score: 1

Accepted answer isThe search continues without reducing the search range, leading to infinite recursion.

20.

Which of the following statements about the program are correct?

1 Point
The recursive calls in Lines 11 and 14 reduce the search space in every recursive step.
The program can correctly perform binary search even if numbers is not sorted.
Line 5 compares the target with every element in the list.
The recursion stops either when the target is found (Line 8) or when the search range becomes invalid (Line 2).

Yes, the answer is correct. Score: 1

Accepted answer isThe recursive calls in Lines 11 and 14 reduce the search space in every recursive step., The recursion stops either when the target is found (Line 8) or when the search range becomes invalid (Line 2).

Week 7 : Assignment 7

Due date on 2026-09-09, 23:59 IST

The due date for submitting this assignment has passed.

Your last recorded submission was on 2026-08-31, 19:04 IST.

Scenario: Drone Surveillance Route

A disaster management team uses a drone to capture images of a disaster-affected area. The images are stored in a matrix. During analysis, the system processes the images in a spiral order, beginning from the outer boundary and gradually moving toward the center so that all regions are examined systematically.

The following Python program performs the spiral traversal of the matrix.

```
def spiral(m, n, a):
    k = 0
    l = 0

    while k < m and l < n:

        for i in range(l, n):
            print(a[k][i], end=" ")
        k += 1

        for i in range(k, m):
            print(a[i][n-1], end=" ")
        n -= 1

        if k < m:
            for i in range(n-1, l-1, -1):
                print(a[m-1][i], end=" ")
            m -= 1

        if l < n:
            for i in range(m-1, k-1, -1):
                print(a[i][l], end=" ")
            l += 1

matrix = [
    [1, 2, 3, 4],
    [5, 6, 7, 8],
    [9, 10, 11, 12],
    [13, 14, 15, 16]
]

spiral(4, 4, matrix)
```

1.

Which variables store the starting row and starting column of the current spiral layer?

1 Point

k and l

m and n

i and k

a and i

Yes, the answer is correct. Score: 1

Accepted answer isk and l

2.

What is the purpose of the statement at line 9?

1 Point

It moves to the next column.

It moves to the next row after printing the top row.

It moves to the previous row.

It resets the row index.

Yes, the answer is correct. Score: 1

Accepted answer isIt moves to the next row after printing the top row.

3.

Which statement best describes the loop at lines 11–12?

1 Point

It prints the first column from top to bottom.

It prints the last row from left to right.

It prints the last column from top to bottom.

It prints the top row from right to left.

Yes, the answer is correct. Score: 1

Accepted answer isIt prints the last column from top to bottom.

4.

What is the first sequence of numbers printed by the program?

1 Point

1 2 3 4

4 8 12 16

13 14 15 16

1 5 9 13

Yes, the answer is correct. Score: 1

Accepted answer is1 2 3 4

5.

Which of the following statements are true about the program?

1 Point

The variables k, l, m, and n define the current boundaries of the unvisited portion of the matrix.

After each side of the spiral is printed, the corresponding boundary is updated.

The program prints only square matrices correctly.

The program traverses the matrix column by column throughout.

Yes, the answer is correct. Score: 1
Accepted answer isThe variables k, l, m, and n define the current boundaries of the unvisited portion of the matrix., After each side of the spiral is printed, the corresponding boundary is updated.

Scenario: Mini Snake and Ladder Tournament

A board game club is testing a simplified Snake and Ladder game. The board has 25 positions. A player starts at position 1 and moves according to a sequence of dice rolls. If the player lands on the bottom of a ladder, the player climbs immediately. If the player lands on the head of a snake, the player slides down immediately. If the player's position becomes greater than 25, the move is ignored.
The game has the following snakes and ladders:

- Ladder: 4 → 12
- Ladder: 9 → 18
- Snake: 16 → 7
- Snake: 22 → 14

The following Python program simulates the player's journey.

```
def play_game():
    position = 1
    target = 25
    rolls = [3, 5, 7, 6, 4, 2]

    for roll in rolls:
        if position + roll <= target:
            position += roll

        if position == 4:
            position = 12
        elif position == 9:
            position = 18
        elif position == 16:
            position = 7
        elif position == 22:
            position = 14

        print(position, end=" ")

        if position == target:
            break

play_game()
```

6.

Which statement ensures that the player does not move beyond the last position on the board?

1 Point

if position == target:
if position + roll <= target:
position += roll
break

Yes, the answer is correct. Score: 1

Accepted answer isif position + roll <= target:

7.

If the player reaches position 16, what happens next?

1 Point

The player moves to position 22.
The player stays at position 16.

The player moves to position 18.
The player moves to position 7.

Yes, the answer is correct. Score: 1

Accepted answer isThe player moves to position 7.

8.

What is the purpose of the statement at line 21?

1 Point

It checks whether all dice rolls are completed.
It checks whether the player has reached the target position.
It resets the player's position.
It moves the player to the next turn.

Yes, the answer is correct. Score: 1

Accepted answer isIt checks whether the player has reached the target position.

9.

Which statement immediately ends the loop once the player reaches the target?

1 Point

break
position += roll
print(position, end=" ")
for roll in rolls:

Yes, the answer is correct. Score: 1

Accepted answer isbreak

10.

Which of the following statements are true about the program?

1 Point

The player always moves after every dice roll.
The game stops only after all dice rolls are processed.
A move is ignored if it takes the player beyond position 25.
Landing on a snake or ladder changes the player's position immediately.

Yes, the answer is correct. Score: 1

Accepted answer isA move is ignored if it takes the player beyond position 25., Landing on a snake or ladder changes the player's position immediately.

Scenario: Food Delivery Pickup Locations

A food delivery company uses a map to display the locations of restaurants where delivery agents need to collect orders. Each restaurant is represented by a marker on the map. When a marker is clicked, the restaurant's name is displayed.

The following Python program creates a map and places markers for three restaurants.

```
1   import gmplot
2
3   gmap = gmplot.GoogleMapPlotter(15.3647, 75.1240, 13)
4
5   restaurants = [
6       ("Pizza Hub", 15.3655, 75.1258),
7       ("Green Bowl", 15.3628, 75.1219),
8       ("Spice Corner", 15.3672, 75.1285)
9   ]
10
11  for name, lat, lon in restaurants:
12      gmap.marker(lat, lon, title=name)
13
14  gmap.draw("restaurants.html")
```

11.

What is the purpose of the statement at line 3?

1 Point

It creates a marker on the map.
It stores restaurant names.
It creates a Google map centered at the given location.
It saves the map as an HTML file.

Yes, the answer is correct. Score: 1

Accepted answer isIt creates a Google map centered at the given location.

12.

Which statement adds a marker for each restaurant?

1 Point

gmap.draw("restaurants.html")
gmap.marker(lat, lon, title=name)
restaurants = [...]
for name, lat, lon in restaurants:

Yes, the answer is correct. Score: 1

Accepted answer isgmap.marker(lat, lon, title=name)

13.

What is displayed as the title of each marker?

1 Point

The latitude value
The longitude value
The map's zoom level
The restaurant name

Yes, the answer is correct. Score: 1

Accepted answer isThe restaurant name

14.

What is the purpose of the statement at line 14?

1 Point

It creates another marker.
It removes all markers from the map.
It changes the map center.
It saves the map as an HTML file.

Yes, the answer is correct. Score: 1

Accepted answer isIt saves the map as an HTML file.

15.

Which of the following statements are true about the program?

1 Point

The program creates a separate map for each restaurant.
The markers are added using the latitude and longitude values.
The program iterates through all restaurants using a for loop.
The HTML file is created before the markers are added.

Yes, the answer is correct. Score: 1

Accepted answer isThe markers are added using the latitude and longitude values., The program iterates through all restaurants using a for loop.

Case Study: Smart Heritage City Explorer

The tourism department of a heritage city has developed a Smart Heritage City Explorer to help visitors explore popular attractions.

The city is represented as a matrix, where each cell corresponds to a location. Every morning, an inspection drone scans the city in a spiral order, starting from the outer boundary and moving towards the center to ensure that every location is inspected exactly once.

Information about each attraction is stored in a dictionary containing the attraction's name, category, visitor rating, latitude, and longitude. When a visitor searches for an attraction, the application uses linear search if the attractions are stored in random order. If the attractions are sorted alphabetically, the application uses binary search to locate the attraction more efficiently.

After an attraction is found, its latitude and longitude are used to place a marker on a Google Map so that visitors can easily navigate to the destination.
The application also includes a Treasure Trail game. Visitors move across 25 checkpoints using dice rolls. Some checkpoints contain ladders that move visitors ahead, while others

contain snakes that move them backward. If a dice roll takes a visitor beyond the final checkpoint, that move is ignored.

The developers use appropriate data structures and algorithms so that the application remains efficient and easy to maintain.

16.

The attractions are stored in random order. Which searching technique is most appropriate for finding a particular attraction?

1 Point

Binary Search
Spiral Search
Linear Search
Depth First Search

Yes, the answer is correct. Score: 1

Accepted answer isLinear Search

17.

Why does the inspection drone use a spiral traversal of the city matrix?

1 Point

To visit only the corner locations.
To inspect every location by moving from the outer boundary towards the center.
To sort the locations alphabetically.
To search for a specific attraction.

Yes, the answer is correct. Score: 1

Accepted answer isTo inspect every location by moving from the outer boundary towards the center.

18.

Which information is essential for placing an attraction on the Google Map?

1 Point

Name and Category
Rating and Category
Latitude and Longitude
Name and Rating

Yes, the answer is correct. Score: 1

Accepted answer isLatitude and Longitude

19.

In the Treasure Trail game, what should happen if a dice roll moves the visitor beyond checkpoint 25?

1 Point

The visitor returns to checkpoint 1.
The visitor wins immediately.
The visitor moves to checkpoint 25.
The move is ignored.

Yes, the answer is correct. Score: 1

Accepted answer isThe move is ignored.

20.

Which of the following statements are true about the Smart Heritage City Explorer?

1 Point

Binary search can be used efficiently even if the attractions are stored in random order.
A dictionary is suitable for storing multiple details about each attraction.
Spiral traversal is a suitable approach for visiting every location in a matrix.
Google Map markers are placed using the latitude and longitude of the visitor rating.

Yes, the answer is correct. Score: 1

Accepted answer isA dictionary is suitable for storing multiple details about each attraction., Spiral traversal is a suitable approach for visiting every location in a matrix.

Week 8 : Assignment 8

Due date on 2026-09-16, 23:59 IST

The due date for submitting this assignment has passed.

Your last recorded submission was on 2026-09-07, 22:11 IST.

Scenario: Student Record Management

A college stores each student's information as a tuple because these details do not change during a semester.

One student's record is stored as:

("Ananya", 1023, "Computer Science", 8.7)

The information is stored in the following order:

- Student Name
- Student ID
- Department
- CGPA

The college uses this tuple whenever it needs to retrieve a student's information.

1.

Which piece of information is stored in the third position (second index) of the tuple?

1 Point

Student Name
Student ID
Department
CGPA

Yes, the answer is correct. Score: 1

Accepted answer isDepartment

2.

Why is a tuple a suitable data structure for storing this record?

1 Point

Because the student details are not expected to change frequently.
Because tuples automatically sort the data.
Because tuples allow duplicate student IDs.
Because tuples can only store text values.

Yes, the answer is correct. Score: 1

Accepted answer isBecause the student details are not expected to change frequently.

3.

How many pieces of information are stored in the student's tuple?

1 Point

2
3
4
5

Yes, the answer is correct. Score: 1

Accepted answer is4

4.

Which of the following is the last item stored in the tuple?

1 Point

Ananya

1023

Computer Science

8.7

Yes, the answer is correct. Score: 1

Accepted answer is8.7

5.

A staff member wants to replace the department "Computer Science" with "Artificial Intelligence" in the existing tuple. What should they do?

1 Point

Modify the existing tuple directly.

Create a new tuple with the updated value.

Delete only the department value.

Use the append() operation.

Yes, the answer is correct. Score: 1

Accepted answer isCreate a new tuple with the updated value.

Scenario: Lucky Draw Betting Game

A gaming café has introduced a simple Lucky Draw Challenge. A player starts with ₹0 in their game account.

The game is played 7 times. In every round:

- The player's bet number is randomly chosen between 1 and 10.
- The lucky draw number is also randomly chosen between 1 and 10.
- If both numbers are the same, the player wins ₹900, but ₹100 is deducted as the participation fee. Hence, the account increases by ₹800.
- If the numbers are different, the player only loses the ₹100 participation fee.
- After every round, the current account balance is displayed.

The following Python program implements this game.

```
1. import random
2.
3. account = 0
4.
5. for i in range(7):
6.     bet = random.randint(1, 10)
7.     lucky_draw = random.randint(1, 10)
8.     print("Bet:", bet)
9.     print("Lucky draw:", lucky_draw)
10.
11.    if bet == lucky_draw:
12.        account = account + 900 - 100
13.    else:
14.        account = account - 100
15.
16.    print("Amount in your game account:", account)
```

6.

According to Line 5, how many rounds will the game be played?

1 Point

7

8

10

Depends on the random numbers

Yes, the answer is correct. Score: 1

Accepted answer is7

7.

Which lines generate the random numbers for the player's bet and the lucky draw?

1 Point

Lines 3 and 5

Lines 8 and 9

Lines 11 and 12

Lines 6 and 7

No, the answer is incorrect. Score: 0

Accepted answer isLines 6 and 7

8.

If the bet number and lucky draw number are equal, what is the net change in the account balance according to Line 12?

1 Point

+₹100
+₹900
+₹700
+₹800

Yes, the answer is correct. Score: 1

Accepted answer is+₹800

9.

According to Lines 13–14, what happens when the bet number and lucky draw number are different?

1 Point

₹100 is deducted from the account.
₹900 is added to the account.
The account balance remains unchanged.
A new account is created.

Yes, the answer is correct. Score: 1

Accepted answer is₹100 is deducted from the account.

10.

Which of the following statements about the program are correct?

1 Point

The player's account balance is printed after every round.
The player's bet is entered by the user using input().
The lucky draw number is always different from the bet number.
The player's account balance can become negative.

Yes, the answer is correct. Score: 1

Accepted answer isThe player's account balance is printed after every round., The player's account balance can become negative.

Scenario: Secure Access Puzzle Generator

A cybersecurity training platform uses anagrams to create authentication puzzles. Before granting access to a practice lab, the system displays a shuffled version of a secret keyword. To ensure the puzzle is meaningful, the displayed word must not be identical to the original keyword. The system keeps generating new arrangements until a valid anagram is

produced. The following Python program implements this logic.

```
1. import random
2.
3. keyword = input("Enter the secret keyword: ")
4. letters = list(keyword)
5.
6. while True:
7.     random.shuffle(letters)
8.     candidate = "".join(letters)
9.
10.    if candidate != keyword:
11.        break
12.
13. print("Authentication Puzzle:", candidate)
```

11.

Suppose the user enters the word LEVEL. Which of the following statements is always true?

1 Point

The displayed puzzle will never contain repeated letters.
The displayed puzzle will always have exactly five letters.
The displayed puzzle will always be alphabetically sorted.
The displayed puzzle cannot be generated because the word contains repeated letters.

Yes, the answer is correct. Score: 1

Accepted answer isThe displayed puzzle will always have exactly five letters.

12.

Consider the execution of Lines 6–11. Which situation can cause the loop to execute more than once?

1 Point

The entered keyword has fewer than five letters.
random.shuffle() arranges the letters in the same order as the original keyword.
The keyword contains only uppercase letters.
The keyword contains duplicate characters.

Yes, the answer is correct. Score: 1

Accepted answer israndom.shuffle() arranges the letters in the same order as the original keyword.

13.

Which of the following inputs may cause the program to never terminate?

1 Point

PYTHON
RANDOM

A
SECURE

Yes, the answer is correct. Score: 1

Accepted answer isA

14.

Which statement best describes the purpose of Lines 10–11?

1 Point

They ensure that every possible anagram is generated exactly once.
They ensure that duplicate letters are removed.
They stop the loop only after a shuffled word different from the original has been generated.
They guarantee that the generated anagram is alphabetically smallest.

Yes, the answer is correct. Score: 1

Accepted answer isThey stop the loop only after a shuffled word different from the original has been generated.

15.

Which of the following statements about the program are correct?

1 Point

If the input contains only one character, the program may run indefinitely.
Every execution of the program produces a different anagram.
The program guarantees that all possible anagrams will eventually be generated.
The generated puzzle always contains exactly the same characters as the original keyword.

Yes, the answer is correct. Score: 1

Accepted answer isIf the input contains only one character, the program may run indefinitely., The generated puzzle always contains exactly the same characters as the original keyword.

Scenario: Customer Feedback Analysis

An online shopping company collects customer reviews after every purchase. The data is stored in a Pandas DataFrame. To understand customer satisfaction, the company uses NLTK's VADER (Valence Aware Dictionary and sEntiment Reasoner) sentiment analyzer. Each review is assigned a compound sentiment score. Reviews with a positive compound score are labeled Positive; otherwise, they are labeled Negative.

The following Python program performs this analysis.

```
1. import pandas as pd
2. from nltk.sentiment.vader import SentimentIntensityAnalyzer
3.
4. reviews = pd.DataFrame({
5.     "Customer": ["Asha", "Rahul", "Neha", "Arjun"],
6.     "Review": [
7.         "Excellent product and fast delivery",
8.         "The quality is average",
9.         "Very disappointed with the service",
10.        "Worth every penny"
11.    ]
12. })
13.
14. analyzer = SentimentIntensityAnalyzer()
15.
16. def get_sentiment(review):
17.     score = analyzer.polarity_scores(review)["compound"]
18.     if score > 0:
19.         return "Positive"
20.     else:
21.         return "Negative"
22.
23. reviews["Sentiment"] = reviews["Review"].apply(get_sentiment)
24.
25. print(reviews)
```

16.

What is the primary purpose of Line 14?

1 Point

To create a Pandas DataFrame.
To create an object that can analyze the sentiment of text.
To read customer reviews from a file.
To display the final DataFrame.

Yes, the answer is correct. Score: 1

Accepted answer isTo create an object that can analyze the sentiment of text.

17.

According to Line 17, which value is used to determine whether a review is positive or negative?

1 Point

neg

compound
neu
pos

Yes, the answer is correct. Score: 1

Accepted answer iscompound

18.

What is the purpose of Line 23?

1 Point

It removes reviews with negative sentiment.
It creates a new column containing the sentiment label for each review.
It sorts the reviews based on sentiment.
It replaces the original review text with sentiment scores.

Yes, the answer is correct. Score: 1

Accepted answer isIt creates a new column containing the sentiment label for each review.

19.

Which statement best describes the role of the function get_sentiment()?

1 Point

It converts reviews into uppercase.
It counts the number of words in each review.
It calculates the average sentiment score of all reviews.
It assigns a sentiment label to a single review based on its compound score.

Yes, the answer is correct. Score: 1

Accepted answer isIt assigns a sentiment label to a single review based on its compound score.

20.

Which of the following statements are correct?

1 Point

The program stores the sentiment score directly in the Review column.
The reviews are stored in a Pandas DataFrame.
The sentiment label for each review is generated using the get_sentiment() function.
The program classifies reviews based on the length of the review text.

Yes, the answer is correct. Score: 1

Accepted answer isThe reviews are stored in a Pandas DataFrame., The sentiment label for each review is generated using the get_sentiment() function.

Week 9 : Assignment 9

Due date on 2026-09-23, 23:59 IST

The due date for submitting this assignment has passed.

Your last recorded submission was on 2026-09-13, 23:17 IST.

Scenario: Product Review Analysis

An online marketplace wants to analyze the words used in customer reviews. The program tokenizes the review text, creates a frequency distribution using FreqDist, and displays only the words that occur more than once. It also reports the total number of unique words.

```
1. from nltk.tokenize import word_tokenize
2. from nltk import FreqDist
3.
4. reviews = """
5. Fast delivery and good packaging.
6. Good quality and fast service.
7. Delivery was fast and packaging was good.
8. """
9.
10. tokens = word_tokenize(reviews.lower())
11. freq = FreqDist(tokens)
12.
13. print(len(tokens))
14. print(len(freq))
15.
16. for word in sorted(freq):
17.     if freq[word] > 1:
18.         print(word, freq[word])
```

1.

Which statement converts the review text into a list of tokens?

1 Point

Line 10
Line 11
Line 13
Line 16

Yes, the answer is correct. Score: 1

Accepted answer isLine 10

2.

What does the expression len(freq) in Line 14 return?

1 Point

Total number of characters in the review
Number of unique tokens in the review
Total number of repeated words
Number of lines in the review

Yes, the answer is correct. Score: 1

Accepted answer isNumber of unique tokens in the review

3.

Which line creates the frequency distribution of the tokens?

1 Point

Line 10
Line 16
Line 11
Line 13

No, the answer is incorrect. Score: 0

Accepted answer isLine 11

4.

Which part of the program prints only the words that occur more than once?

1 Point

Line 10
Line 13
Line 14
Lines 16–18

Yes, the answer is correct. Score: 1

Accepted answer isLines 16–18

5.

Which of the following statements are correct?

1 Point

len(tokens) returns the number of unique words in the review.
word_tokenize() automatically counts the frequency of each token.
FreqDist stores each token as a key and its frequency as the corresponding value.
len(freq) gives the number of distinct tokens present in the text.

Partially Correct. Score: 0.5

Accepted answer isFreqDist stores each token as a key and its frequency as the corresponding value., len(freq) gives the number of distinct tokens present in the text.

Scenario: Library Book Recommendation Network

A digital library recommends books based on their similarity. Each book is represented as a node, and an edge between two books indicates that they are frequently borrowed together. The librarian uses NetworkX to model and examine this recommendation network.

NPTE

6.

In this graph, what does a node represent?

1 Point

A book

A recommendation between two books

The total number of books

A borrowing record

Yes, the answer is correct. Score: 1

Accepted answer isA book

7.

What does an edge between two books indicate?

1 Point

The books have the same author.

The books belong to different categories.

The books are borrowed together and are recommended to each other.

The books have the same publication year.

Yes, the answer is correct. Score: 1

Accepted answer isThe books are borrowed together and are recommended to each other.

8.

What does len(G.nodes) represent?

1 Point

Number of recommendations

Number of books in the network

Number of neighboring books

Number of connected components

Yes, the answer is correct. Score: 1

Accepted answer isNumber of books in the network

9.

Why is G.neighbors(book) used in the program?

1 Point

To remove duplicate books

To sort the books alphabetically

To calculate the shortest path

To find books directly connected to a given book

Yes, the answer is correct. Score: 1

Accepted answer isTo find books directly connected to a given book

10.

Which of the following statements are correct?

1 Point

Every node must be connected to every other node.

An edge can exist without any nodes.

len(G.edges) returns the total number of recommendation links in the graph.
A graph can be used to model relationships between connected objects.

Yes, the answer is correct. Score: 1

Accepted answer islen(G.edges) returns the total number of recommendation links in the graph., A graph can be used to model relationships between connected objects.

Scenario: Six Degrees of Separation

A social networking platform represents its users as a graph. Each person is a node, and a friendship between two people is represented by an edge. According to the "Six Degrees of Separation" concept, any two people in a social network can usually be connected through a small chain of mutual friends. The platform analyzes these connections to recommend new friends and understand how closely users are connected.

11.

In the social network described above, what does an edge represent?

1 Point

A user profile
A friendship between two users
The total number of users
A friend recommendation

Yes, the answer is correct. Score: 1

Accepted answer isA friendship between two users

12.

Alice is friends with Bob, and Bob is friends with Carol. Alice and Carol are not directly connected. What exists between Alice and Carol?

1 Point

A path through Bob
A duplicate edge
A disconnected graph
A self-loop

Yes, the answer is correct. Score: 1

Accepted answer isA path through Bob

13.

Why is the shortest path between two users often important in a social network?

1 Point

It increases the total number of users.
It removes duplicate friendships.
It identifies the minimum number of friendship links connecting two users.
It converts the graph into a directed graph.

Yes, the answer is correct. Score: 1

Accepted answer isIt identifies the minimum number of friendship links connecting two users.

14.

When is a social network considered to be connected?

1 Point
Every user has exactly one friend.
Every user is directly connected to every other user.
The graph has no edges.
A path exists between every pair of users.

Yes, the answer is correct. Score: 1

Accepted answer isA path exists between every pair of users.

15.

Which of the following statements are correct?

1 Point
Every pair of users must have a direct friendship for a path to exist.
A graph cannot have more than one path between two users.
A node represents an individual user in the social network.
The Six Degrees of Separation concept is based on finding paths between people in a network.

Yes, the answer is correct. Score: 1

Accepted answer isA node represents an individual user in the social network., The Six Degrees of Separation concept is based on finding paths between people in a network.

Scenario: Quality Inspection of Product Images

An online retail company inspects product images before publishing them on its website. The system opens an image, checks the color of a specific pixel, marks that pixel with a new color for verification, creates a small thumbnail image, and saves the modified image.

```
1. from PIL import Image
2.
3. img = Image.open("product.jpg")
4.
5. print(img.size)
6. print(img.getpixel((20, 20)))
7.
8. img.putpixel((20, 20), (255, 0, 0))
9.
10. thumb = Image.new("RGB", (100, 100), "white")
11. thumb.save("thumbnail.jpg")
12.
13. img.save("verified_product.jpg")
```

16.

Why is getpixel() used in this program?

1 Point

To resize the image
To change the image dimensions
To retrieve the color value of a specific pixel
To save the image

Yes, the answer is correct. Score: 1

Accepted answer isTo retrieve the color value of a specific pixel

17.

What is the purpose of putpixel()?

1 Point

To create a new image
To modify the color of a specific pixel
To compress the image
To display the image

Yes, the answer is correct. Score: 1

Accepted answer isTo modify the color of a specific pixel

18.

What is the purpose of Image.new()?

1 Point

To open an existing image
To copy an existing image
To convert an image into grayscale
To create a new blank image

Yes, the answer is correct. Score: 1

Accepted answer isTo create a new blank image

19.

Why is the save() method used?

1 Point

To store the processed image as a file
To display the image on the screen
To remove unwanted pixels
To count the number of pixels

Yes, the answer is correct. Score: 1

Accepted answer isTo store the processed image as a file

20.

Which of the following statements are correct?

1 Point

getpixel() changes the color of a pixel.
Image.new() creates a new image with the specified mode and dimensions.
putpixel() automatically saves the modified image to disk.
Changes made using putpixel() are stored in memory until the image is saved.

Yes, the answer is correct. Score: 1

Accepted answer isImage.new() creates a new image with the specified mode and dimensions., Changes made using putpixel() are stored in memory until the image is saved.

Week 10 : Assignment 10

Due date on 2026-09-30, 23:59 IST

The due date for submitting this assignment has passed.

Your last recorded submission was on 2026-09-19, 08:29 IST.

Scenario - FLAMES

A school is conducting a fun activity where students enter two names to predict their relationship using the classic FLAMES game. The program removes all common letters from the two names, counts the remaining letters, and repeatedly eliminates letters from the word FLAMES until only one letter remains. Based on the final letter, the corresponding relationship is displayed.

Study the program below and answer the questions.

```
1. name1 = input("Enter first name: ").lower().replace(" ", "")
2. name2 = input("Enter second name: ").lower().replace(" ", "")

3. list1 = list(name1)
4. list2 = list(name2)

5. for ch in name1:
6.     if ch in list2:
7.         list1.remove(ch)
8.         list2.remove(ch)

9. count = len(list1) + len(list2)

10. flames = ["F", "L", "A", "M", "E", "S"]

11. while len(flames) > 1:
12.     index = (count - 1) % len(flames)
13.     flames.pop(index)
14.     flames = flames[index:] + flames[:index]

15. result = {
16.     "F": "Friends",
17.     "L": "Love",
18.     "A": "Affection",
19.     "M": "Marriage",
20.     "E": "Enemies",
21.     "S": "Siblings"
22. }

23. print(result[flames[0]])
```

1.

What is the purpose of the statements in Lines 1 and 2?

1 Point

Convert both names to uppercase.
Convert names to lowercase and remove spaces.
Reverse both names.
Remove duplicate letters from the names.

Yes, the answer is correct. Score: 1

Accepted answer isConvert names to lowercase and remove spaces.

2.

What is stored in the variable count at Line 9?

1 Point

Total number of letters in the original names.
Number of common letters.
Total number of remaining letters after removing common letters.
Number of letters in the word FLAMES.

Yes, the answer is correct. Score: 1

Accepted answer isTotal number of remaining letters after removing common letters.

3.

Which statement actually removes a letter from the FLAMES list during each iteration?

1 Point

Line 11
Line 12
Line 13
Line 14

Yes, the answer is correct. Score: 1

Accepted answer isLine 13

4.

Why is the modulo operator (%) used in Line 12?

1 Point

To convert the index into a floating-point value.
To ensure the index always stays within the current size of the list.
To sort the FLAMES list.
To count the number of common letters.

Yes, the answer is correct. Score: 1

Accepted answer isTo ensure the index always stays within the current size of the list.

5.

Which of the following statements are correct about the program?

1 Point

The loop in Lines 11–14 continues until only one letter remains in the flames list.
The dictionary result maps each FLAMES letter to a relationship.
The program removes all vowels before calculating the result.
list(name1) in Line 3 converts the first name into a list of characters.

Yes, the answer is correct. Score: 1

Accepted answer isThe loop in Lines 11–14 continues until only one letter remains in the flames list., The dictionary result maps each FLAMES letter to a relationship., list(name1) in Line 3 converts the first name into a list of characters.

Scenario – FLAMES 2

A software company has developed a Team Compatibility Analyzer to compare the names of two employees working on a project. The program removes the common letters from both names, counts the remaining letters, and repeatedly eliminates options from a predefined list until only one category remains. The final category is then displayed as the team's compatibility result.

Study the program carefully and answer the following questions.

```
1. name1 = input("Enter first name: ").lower().replace(" ", "")
2. name2 = input("Enter second name: ").lower().replace(" ", "")

3. list1 = list(name1)
4. list2 = list(name2)

5. for ch in name1:
6.     if ch in list2:
7.         list1.remove(ch)
8.         list2.remove(ch)

9. count = len(list1) + len(list2)

10. flames = ["F", "L", "A", "M", "E", "S"]

11. while len(flames) > 1:
12.     index = (count - 1) % len(flames)
13.     flames.pop(index)
14.     flames = flames[index:] + flames[:index]

15. result = {
16.     "F": "Friendly",
17.     "L": "Leader-Follower",
18.     "A": "Adaptive",
19.     "M": "Mentor-Mentee",
20.     "E": "Efficient",
21.     "S": "Supportive"
22. }

23. print(result[flames[0]])
```

6.

Why are the methods lower() and replace(" ", "") used in Lines 1 and 2 before comparing the names?

1 Point

To sort the letters in each name.
To make the comparison independent of letter case and spaces.
To remove duplicate characters.
To reverse the names before processing.

Yes, the answer is correct. Score: 1

Accepted answer isTo make the comparison independent of letter case and spaces.

7.

Which statement correctly describes the value stored in count at Line 9?

1 Point

The total number of characters entered by the user.
The number of matching characters between the two names.
The total number of characters remaining after common letters are removed.
The number of elements in the flames list.

Yes, the answer is correct. Score: 1

Accepted answer isThe total number of characters remaining after common letters are removed.

8.

Which line changes the size of the flames list during each iteration of the loop?

1 Point

Line 11
Line 12
Line 13
Line 14

Yes, the answer is correct. Score: 1

Accepted answer isLine 13

9.

Suppose count is greater than the current length of the flames list. Why does the program still work correctly?

1 Point

Python automatically increases the size of the list.
The modulo operator (%) converts the value into a valid list index.
The list is sorted before accessing an element.
The while loop ignores values larger than the list size.

Yes, the answer is correct. Score: 1

Accepted answer isThe modulo operator (%) converts the value into a valid list index.

10.

Which of the following statements about the program are correct?

1 Point

The while loop continues until only one element remains in the flames list.
Removing Line 14 would change the sequence in which elements are eliminated.

The dictionary result is used to remove elements from the flames list.
The value of count is computed only after common letters have been removed from both names

Yes, the answer is correct. Score: 1

Accepted answer isThe while loop continues until only one element remains in the flames list., Removing Line 14 would change the sequence in which elements are eliminated., The value of count is computed only after common letters have been removed from both names

Scenario: Wild Life

A wildlife research team is studying drone images of a forest to estimate the amount of sunlight reaching the ground. They use a Python program that examines every pixel in an image. For each pixel, the program calculates the average intensity of its Red (R), Green (G), and Blue (B) values. If the average intensity is greater than 127, the pixel is classified as Bright; otherwise, it is classified as Dark. Finally, the program displays the total number of bright and dark pixels in the image.

Study the following program and answer the questions.

```
1. from PIL import Image

2. img = Image.open("sample.jpg")

3. pixels = img.load()

4. width, height = img.size

5. bright = 0
6. dark = 0

7. for x in range(width):
8.     for y in range(height):
9.         r, g, b = pixels[x, y]

10.         average = (r + g + b) // 3

11.         if average > 127:
12.             bright += 1
13.         else:
14.             dark += 1

15. print("Bright Pixels:", bright)
16. print("Dark Pixels:", dark)
```

11.

What is the purpose of Line 3 (pixels = img.load())?

1 Point

It opens the image file.
It creates a pixel access object to read individual pixel values.
It converts the image into grayscale.
It returns the width and height of the image.

Yes, the answer is correct. Score: 1

Accepted answer isIt creates a pixel access object to read individual pixel values.

12.

What is stored in the variables width and height at Line 4?

1 Point

The total number of pixels in the image.
The RGB values of the first pixel.
The dimensions of the image in pixels.
The brightness and darkness counts.

Yes, the answer is correct. Score: 1

Accepted answer isThe dimensions of the image in pixels.

13.

What is the value of r, g, and b after executing Line 9?

1 Point

The row, column, and pixel number.
The Red, Green, and Blue intensity values of the current pixel.
The width, height, and brightness of the image.
Three randomly generated numbers.

Yes, the answer is correct. Score: 1

Accepted answer isThe Red, Green, and Blue intensity values of the current pixel.

14.

Why is integer division (//) used in Line 10?

1 Point

To compute the average RGB intensity as a whole number.
To increase the brightness of every pixel.
To convert the image into grayscale.
To avoid using nested loops.

Yes, the answer is correct. Score: 1

Accepted answer isTo compute the average RGB intensity as a whole number.

15.

Suppose an image has 200 × 100 pixels. How many times will the statement in Line 9 execute?

1 Point

200
100
300
20,000

Yes, the answer is correct. Score: 1
Accepted answer is20,000

Scenario: PixelMap Image Quality Inspector
A photography studio has developed a PixelMap Image Quality Inspector to identify overexposed regions in digital photographs. The program scans every pixel in an image using the PixelMap, calculates the average intensity of its Red (R), Green (G), and Blue (B) values, and classifies each pixel as Bright or Dark based on a threshold value. At the end, it reports the total number of bright and dark pixels.
Study the following program and answer the questions.

```python
from PIL import Image

img = Image.open("sample.jpg")

pixels = img.load()

width, height = img.size

bright = 0
dark = 0

for x in range(width):
    for y in range(height):
        r, g, b = pixels[x, y]

        average = (r + g + b) // 3

        if average > 127:
            bright += 1
        else:
            dark += 1

print("Bright Pixels:", bright)
print("Dark Pixels:", dark)
```

16.

Which statement ensures that every pixel in the image is processed exactly once?

1 Point
Line 3
Lines 7–8
Line 10
Line 15

Yes, the answer is correct. Score: 1

Accepted answer isLines 7–8

17.

What is the data type of the value returned by pixels[x, y] for a color image?

1 Point
Integer
String
Tuple
List

Yes, the answer is correct. Score: 1

Accepted answer isTuple

18.

If the current pixel has RGB values (90, 150, 180), what will be the value of average?

1 Point
120
140
150
420

Yes, the answer is correct. Score: 1

Accepted answer is140

19.

If a pixel has an average intensity of 127, how will it be classified?

1 Point
Bright
Dark
Both Bright and Dark
The program generates an error.

Yes, the answer is correct. Score: 1

Accepted answer isDark

20.

Which of the following statements are correct?

1 Point
img.load() allows the program to access individual pixel values.
The variables bright and dark are used as counters.
The nested loops execute once for each row only.
The program processes pixels using their (x, y) coordinates.

Yes, the answer is correct. Score: 1

Accepted answer isimg.load() allows the program to access individual pixel values., The variables bright and dark are used as counters., The program processes pixels using their (x, y) coordinates.

Week 11 : Assignment 11

Due date on 2026-10-07, 23:59 IST

The due date for submitting this assignment has passed.

Your last recorded submission was on 2026-09-27, 11:20 IST.

Scenario: Semester Planner

A faculty coordinator is preparing the academic schedule for a semester. The coordinator wants to display the calendar of a selected month and identify the weekday on which an important event is scheduled. The following Python program is used.

```
1. import calendar
2.
3. year = int(input("Enter the year: "))
4. month = int(input("Enter the month (1-12): "))
5. day = int(input("Enter the event date: "))
6.
7. print("Semester Planner")
8. print("-" * 25)
9. print(calendar.month(year, month))
10.
11. weekday = calendar.weekday(year, month, day)
12. days = ["Monday", "Tuesday", "Wednesday", "Thursday",
13.         "Friday", "Saturday", "Sunday"]
14.
15. print("Event Day:", days[weekday])
```

1.

Which function is used to display the calendar of the selected month?

1 Point

calendar.weekday()
calendar.month()
calendar.date()
calendar.day()

Yes, the answer is correct. Score: 1

Accepted answer iscalendar.month()

2.

What is stored in the variable weekday after Line 11 executes?

1 Point
The name of the weekday
The entered date
The index representing the weekday
The month number

Yes, the answer is correct. Score: 1

Accepted answer isThe index representing the weekday

3.

Why is the list days created in Lines 12–13?

1 Point
To store all months of the year
To convert the weekday index into its corresponding weekday name
To count the number of weeks in the month
To validate the entered date

Yes, the answer is correct. Score: 1

Accepted answer isTo convert the weekday index into its corresponding weekday name

4.

Which statement displays the weekday name of the event?

1 Point
print(calendar.month(year, month))
weekday = calendar.weekday(year, month, day)
print("Semester Planner")
print("Event Day:", days[weekday])

Yes, the answer is correct. Score: 1

Accepted answer isprint("Event Day:", days[weekday])

5.

Which of the following statements about the program are correct?

1 Point
The user enters the year, month, and day.
calendar.weekday() returns the name of the weekday.
The list days is used to obtain the month name from the returned index.
The calendar is displayed before the weekday is printed.

No, the answer is incorrect. Score: 0

Accepted answer isThe user enters the year, month, and day., The calendar is displayed before the weekday is printed.

Scenario: Meeting Schedule Planner

An office administrator is preparing the schedule for monthly team meetings. The administrator wants to display the calendar of a selected month, determine whether the year is a leap year, and count the number of Mondays in that month. The following Python program is used.

```
1. import calendar
2.
3. year = int(input("Enter the year: "))
4. month = int(input("Enter the month (1-12): "))
5.
6. print("Meeting Schedule")
7. print(calendar.month(year, month))
8.
9. if calendar.isleap(year):
10.     print("Leap Year")
11. else:
12.     print("Not a Leap Year")
13.
14. weeks = calendar.monthcalendar(year, month)
15. monday_count = 0
16.
17. for week in weeks:
18.     if week[0] != 0:
19.         monday_count += 1
20.
21. print("Number of Mondays:", monday_count)
```

6.

What is the purpose of calendar.isleap(year)?

1 Point

To display the calendar of the year
To check whether the given year is a leap year
To count the number of days in the month
To find the weekday of the first day of the month

Yes, the answer is correct. Score: 1

Accepted answer isTo check whether the given year is a leap year

7.

What is stored in the variable weeks after Line 14?

1 Point

A list containing the weeks of the month
The number of weeks in the month
A string representing the calendar
The weekday of the first day

Yes, the answer is correct. Score: 1

Accepted answer isA list containing the weeks of the month

8.

Why does the program check if week[0] != 0?

1 Point

To count only the weeks with at least one holiday
To count only the Mondays that exist in the selected month
To skip weekends
To find the first week of the month

Yes, the answer is correct. Score: 1

Accepted answer isTo count only the Mondays that exist in the selected month

9.

Which variable stores the total number of Mondays in the selected month?

1 Point

weeks
month
weekday
monday_count

Yes, the answer is correct. Score: 1

Accepted answer ismonday_count

10.

Which of the following statements are correct?

1 Point

calendar.monthcalendar() returns a list of weeks.
The program counts the number of days in the selected month.
calendar.isleap() returns the number of days in February.
The calendar is displayed before the number of Mondays is printed.

Yes, the answer is correct. Score: 1

Accepted answer iscalendar.monthcalendar() returns a list of weeks., The calendar is displayed before the number of Mondays is printed.

Scenario: Shift Allocation Calendar

A manufacturing company prepares employee shift schedules every month. Before assigning shifts, the HR team wants to display the calendar, determine the weekday of the first working day of the month, and count how many weekends (Saturdays and Sundays) occur during that month. The following Python program is used.

```
1. import calendar
2.
3. year = int(input("Enter the year: "))
4. month = int(input("Enter the month (1-12): "))
5.
6. print("Shift Allocation Calendar")
7. print(calendar.month(year, month))
8.
9. weeks = calendar.monthcalendar(year, month)
10. first_day = calendar.weekday(year, month, 1)
11.
12. day_names = ["Monday", "Tuesday", "Wednesday",
13.              "Thursday", "Friday", "Saturday", "Sunday"]
14.
15. weekend_count = 0
16.
17. for week in weeks:
18.     if week[5] != 0:
19.         weekend_count += 1
20.     if week[6] != 0:
21.         weekend_count += 1
22.
23. print("First day of the month:", day_names[first_day])
24. print("Total weekends:", weekend_count)
```

11.

What is the value stored in first_day after Line 10 executes?

1 Point

The name of the first weekday of the month
The date of the first Monday in the month
An integer representing the weekday of the first day of the month
The total number of weekdays in the month

Yes, the answer is correct. Score: 1

Accepted answer isAn integer representing the weekday of the first day of the month

12.

Why are the conditions week[5] != 0 and week[6] != 0 checked inside the loop?

1 Point

To count Saturdays and Sundays that belong to the selected month
To count all working days in the month
To find the first weekend of the month
To identify public holidays

Yes, the answer is correct. Score: 1

Accepted answer isTo count Saturdays and Sundays that belong to the selected month

13.

Which statement best describes the variable weeks?

1 Point

It stores the names of all weekdays.
It stores a two-dimensional list where each inner list represents one week.
It stores the total number of weeks in the month.
It stores all weekend dates only.

Yes, the answer is correct. Score: 1

Accepted answer isIt stores a two-dimensional list where each inner list represents one week.

14.

If a selected month contains 5 Saturdays and 4 Sundays, what value will be printed for weekend_count?

1 Point

4
5
8
9

Yes, the answer is correct. Score: 1

Accepted answer is9

15.

Which of the following statements are correct?

1 Point

calendar.weekday(year, month, 1) is used to determine the weekday of the first day of the month.
Each element of weeks is a string representing one week.
The program displays the calendar before counting the weekends.
The value of weekend_count can never be greater than 8.

Yes, the answer is correct. Score: 1

Accepted answer iscalendar.weekday(year, month, 1) is used to determine the weekday of the first day of the month., The program displays the calendar before counting the weekends.

Scenario: Training Session Scheduler

A training coordinator is planning employee training sessions for a selected month. The program displays the calendar, identifies all Fridays in the month, and determines the last Friday available for conducting a monthly review meeting.

```
import calendar

year = int(input("Enter the year: "))
month = int(input("Enter the month (1-12): "))

print("Training Session Scheduler")
print(calendar.month(year, month))

weeks = calendar.monthcalendar(year, month)
fridays = []

for week in weeks:
    if week[4] != 0:
        fridays.append(week[4])

print("Fridays:", fridays)

if len(fridays) >= 4:
    print("Last Friday:", fridays[-1])
else:
    print("Insufficient Fridays")
```

16.

What is stored in the list fridays after the loop completes?

1 Point

The weekday numbers of all Fridays
The dates that fall on Fridays in the selected month
The week numbers containing Fridays
The total number of Fridays

Yes, the answer is correct. Score: 1

Accepted answer isThe dates that fall on Fridays in the selected month

17.

Why is the condition week[4] != 0 used?

1 Point

To ignore holidays that fall on Fridays
To ensure the Friday belongs to the selected month
To identify leap years
To count only the first Friday

Yes, the answer is correct. Score: 1

Accepted answer isTo ensure the Friday belongs to the selected month

18.

What is the purpose of fridays[-1] in Line 19?

1 Point

It returns the first Friday of the month.
It returns the total number of Fridays.
It returns the last element in the fridays list.
It removes the last Friday from the list.

Yes, the answer is correct. Score: 1

Accepted answer isIt returns the last element in the fridays list.

19.

Suppose the selected month has Friday dates 3, 10, 17, 24, and 31. What will be printed by Line 19?

1 Point

24
17
5
31

Yes, the answer is correct. Score: 1

Accepted answer is31

20.

Which of the following statements are correct?

1 Point

calendar.monthcalendar() returns a string containing the monthly calendar.
week[4] represents the fifth week of the month.
The list fridays stores the dates that fall on Fridays.
The program displays the last Friday only if at least four Fridays are present in the month.

Partially Correct. Score: 0.5

Accepted answer isThe list fridays stores the dates that fall on Fridays., The program displays the last Friday only if at least four Fridays are present in the month.

Week 12 : Assignment 12

The due date for submitting this assignment has passed.

Your last recorded submission was on 2026-10-02, 18:36 IST.

Scenario: Ranking the Most Influential Campus Clubs

A university has created a student portal where every club has its own webpage. Each club page contains links to other club pages that it collaborates with or recommends. The university wants the homepage to display the most influential clubs first, not simply those with the highest number of incoming links. They decide to use the PageRank algorithm because it considers both the quantity and quality of incoming links. Clubs that receive recommendations from already influential clubs are expected to receive higher rankings. Answer the following questions based on this scenario.

1.

Why is the university using PageRank instead of simply counting the number of incoming links?

1 Point

Because it considers the importance of the pages providing the links.
Because it ranks pages alphabetically.
Because it only counts outgoing links.
Because it ignores linked pages completely.

Yes, the answer is correct. Score: 1

Accepted answer isBecause it considers the importance of the pages providing the links.

2.

Suppose a newly created club page receives a recommendation from a highly ranked club. What is the most likely outcome?

1 Point

The new club will always become the highest-ranked page immediately.
The recommendation has no effect because the page is new.
The new club's PageRank is likely to improve due to the influential incoming link.
The recommending club loses all of its PageRank.

Yes, the answer is correct. Score: 1

Accepted answer isThe new club's PageRank is likely to improve due to the influential incoming link.

3.

If one club page links to hundreds of other club pages, how does this affect the value of each outgoing link?

1 Point

Each outgoing link becomes more valuable.
Each outgoing link shares only a smaller portion of the page's ranking value.
Outgoing links are ignored by PageRank.
The linked pages receive identical final rankings.

Yes, the answer is correct. Score: 1

Accepted answer isEach outgoing link shares only a smaller portion of the page's ranking value.

4.

Which statement best describes how PageRank determines page importance?

1 Point

Pages with the longest content always rank highest.
Pages created first always receive the highest rank.
Pages with the largest file size receive better rankings.
Importance is determined iteratively based on the network of incoming links.

Yes, the answer is correct. Score: 1

Accepted answer isImportance is determined iteratively based on the network of incoming links.

5.

Which of the following statements about PageRank are correct?

1 Point

A link from an influential page generally contributes more than a link from an unimportant page.
The algorithm depends only on the total number of incoming links.
The ranking values are refined over multiple iterations.
Every outgoing link from a page transfers a zero share of that page's ranking value

Yes, the answer is correct. Score: 1

Accepted answer isA link from an influential page generally contributes more than a link from an unimportant page., The ranking values are refined over multiple iterations.

Scenario: Ranking Webpages Using a Simplified PageRank Score

A search engine developer is testing a simplified version of the PageRank algorithm. Each webpage starts with an initial score of 1.0. During one update cycle, every page distributes its score equally among all the pages it links to. The following program computes the contribution received by each page after one iteration.

```
1. pages = ['A', 'B', 'C', 'D']
2. links = {
3.     'A': ['B', 'C'],
4.     'B': ['C'],
5.     'C': ['A', 'B', 'D'],
6.     'D': ['A']
7. }
8.
9. rank = {'A':1.0, 'B':1.0, 'C':1.0, 'D':1.0}
10. new_rank = {'A':0, 'B':0, 'C':0, 'D':0}
11.
12. for page in pages:
13.     share = rank[page] / len(links[page])
14.     for dest in links[page]:
15.         new_rank[dest] += share
16.
17. print(new_rank)
```

Answer the following questions based on the program.

6.

What is the purpose of line 13?

1 Point

It counts the number of webpages.
It divides the page's current score equally among its outgoing links.
It assigns a random score to each page.
It resets the PageRank values.

Yes, the answer is correct. Score: 1

Accepted answer isIt divides the page's current score equally among its outgoing links.

7.

Which page receives the greatest total contribution after this iteration?

1 Point

A
B
C
D

Yes, the answer is correct. Score: 1

Accepted answer isC

8.

Which page contributes the smallest amount to each linked page?

1 Point

A
B
C
D

Yes, the answer is correct. Score: 1

Accepted answer isC

9.

Which dictionary stores the updated PageRank values after one iteration?

1 Point

pages
links
rank
new_rank

Yes, the answer is correct. Score: 1

Accepted answer isnew_rank

10.

Which of the following statements are correct about the program?

1 Point

A page with more outgoing links distributes a larger share to each linked page.
Every page starts with the same initial PageRank value.
The program updates rank directly inside the loop.
The program performs exactly one PageRank iteration.

Yes, the answer is correct. Score: 1
Accepted answer isEvery page starts with the same initial PageRank value., The program performs exactly one PageRank iteration.

Scenario: Evaluating One Iteration of a Simplified PageRank Algorithm

A search engine engineer is analyzing how PageRank values change after one iteration. Every webpage starts with an equal rank of 1.0. During each iteration, a page distributes its rank equally among all the pages it links to. The engineer uses the following program to calculate the updated ranks after a single iteration.

```
1. pages = ['P', 'Q', 'R', 'S']
2.
3. links = {
4.     'P': ['Q', 'R'],
5.     'Q': ['P', 'R', 'S'],
6.     'R': ['S'],
7.     'S': ['P', 'Q']
8. }
9.
10. rank = {page: 1.0 for page in pages}
11. updated = {page: 0 for page in pages}
12.
13. for page in pages:
14.     contribution = rank[page] / len(links[page])
15.     for neighbour in links[page]:
16.         updated[neighbour] += contribution
17.
18. print(updated)
```

Answer the following questions based on the program.

11.

Which page contributes the largest value to each page it links to?

1 Point

P

Q

R

S

Yes, the answer is correct. Score: 1

Accepted answer isR

12.

Which page receives contributions from both P and S?

1 Point

P
Q
R
S

Yes, the answer is correct. Score: 1

Accepted answer isQ

13.

Which page will have the highest updated rank after this iteration?

1 Point

P
Q
R
S

Yes, the answer is correct. Score: 1

Accepted answer isS

14.

Why is a separate dictionary (updated) used instead of modifying rank directly inside the loop?

1 Point

To reduce memory usage.
To ensure every contribution is computed using the original ranks.
To avoid sorting the pages.
To prevent duplicate page names.

Yes, the answer is correct. Score: 1

Accepted answer isTo ensure every contribution is computed using the original ranks.

15.

Which of the following statements are correct?

1 Point

A page with fewer outgoing links transfers a larger share to each linked page.
The rank dictionary is modified during the iteration.
The total contribution distributed by each page is equal to its current rank.
All updated ranks are computed before replacing the original ranks.

Yes, the answer is correct. Score: 1

Accepted answer isA page with fewer outgoing links transfers a larger share to each linked page., The total contribution distributed by each page is equal to its current rank., All updated ranks are computed before replacing the original ranks.

Exploring the Collatz Conjecture

A mathematics club is studying the Collatz Conjecture, which states that for any positive integer:

- If the number is even, divide it by 2.
- If the number is odd, multiply it by 3 and add 1.

The process is repeated until the value becomes 1. A student writes the following Python program to generate the sequence for a given starting number. Answer the following questions based on the program.

```
1. n = int(input())
2. steps = 0
3.
4. while n != 1:
5.     print(n, end=" ")
6.     if n % 2 == 0:
7.         n = n // 2
8.     else:
9.         n = 3 * n + 1
10.    steps += 1
11.
12. print(1)
13. print(steps)
```

16.

If the input is 8, how many times is the statement on line 10 executed?

1 Point

3

4

5

6

No, the answer is incorrect. Score: 0

Accepted answer is3

17.

Which statement correctly describes the purpose of line 6?

1 Point

It checks whether the current number is prime.
It checks whether the current number is even.
It checks whether the sequence has ended.
It checks whether the number is greater than 1.

Yes, the answer is correct. Score: 1

Accepted answer isIt checks whether the current number is even.

18.

If the input is 3, which number is printed immediately after 10?

1 Point

8
16
5
20

Yes, the answer is correct. Score: 1

Accepted answer is5

19.

What is the role of the variable steps?

1 Point

It stores the largest value in the sequence.
It counts the number of even numbers encountered.
It stores the final value of the sequence.
It counts the number of transformations performed before reaching 1.

Yes, the answer is correct. Score: 1

Accepted answer isIt counts the number of transformations performed before reaching 1.

20.

Which of the following statements are correct?

1 Point

The value 1 is printed even though the loop terminates before processing it.
For an odd value of n, the next value is always smaller than the current value.
The variable steps is incremented exactly once during every loop iteration.
The program prints every intermediate value generated by the sequence.

Yes, the answer is correct. Score: 1

Accepted answer isThe value 1 is printed even though the loop terminates before processing it., The variable steps is incremented exactly once during every loop iteration., The program prints every intermediate value generated by the sequence.